Magnesium Compounds

ULLMANN'S ENCYCLOPEDIA OF INDUSTRIAL CHEMISTRY

Margarete Seeger, Didier Werke AG, Wiesbaden, Federal Republic of Germany

Walter Otto, retired Kali und Salz AG, Kassel, Federal Republic of Germany

Wilhelm Flick, Hüls AG, Niederkassel, Federal Republic of Germany

Friedrich Bickelhaupt, Vrije Universiteit, Amsterdam, The Netherlands

Otto S. Akkerman, Vrije Universiteit, Amsterdam, The Netherlands

1. Introduction . . . 41
2. Magnesium Carbonate . . . 42
3. Magnesium Chloride . . . 43
3.1. Properties . . . 43
3.2. Raw Materials . . . 44
3.3. Production . . . 44
3.3.1. Precipitation of Magnesium Hydroxide from Seawater . . . 44
3.3.2. Recovery from Spent Liquors from the Potash Industry . . . 44
3.3.3. Evaporation of Seawater and Natural Brines . . . 45
3.3.4. Solution Mining . . . 47
3.3.5. Other Methods . . . 47
3.3.6. Production of Magnesium Chloride Hydrates . . . 47
3.3.7. Production of Anhydrous Magnesium Chloride . . . 48
3.3.8. Purification of Magnesium Chloride Solutions . . . 48
3.3.9. Materials of Construction for the Evaporation of Magnesium Chloride Solutions . . . 49
3.4. Environmental and Legal Aspects . . . 49
3.5. Quality Specifications and Analysis . . . 49
3.6. Transportation and Storage . . . 50
3.7. Uses . . . 50
3.8. Economic Aspects . . . 50
3.9. Toxicology and Occupational Health . . . 51
4. Magnesium Oxide and Hydroxide . . . 51
4.2. Properties . . . 51
4.3. Production of Magnesium Oxide . . . 52
4.3.1. Production from Magnesite . . . 52
4.3.2. Production from Seawater . . . 55
4.3.3. Production from Brines . . . 56
4.3.4. Miscellaneous Processes . . . 58
4.3.5. Burning of Caustic and Sintered Magnesia . . . 58
4.4. Uses . . . 60
4.4.1. Use of Sintered Magnesia as a Refractory . . . 60
4.4.2. Use of Sintered Magnesia as a Heat-Storage Medium . . . 61
4.4.3. Uses of Caustic Magnesia . . . 62
4.4.4. Uses of Magnesium Hydroxide . . . 63
4.5. Economic Aspects . . . 63
4.6. Fused Magnesium Oxide . . . 63
4.6.1. Properties . . . 63
4.6.2. Production . . . 64
4.6.3. Uses . . . 65
4.6.4. Economic Aspects . . . 66
5. Magnesium Sulfate . . . 66
5.1. Properties . . . 66
5.2. Occurrence and Raw Materials . . . 67
5.3. Production . . . 68
5.3.1. Production of Kieserite ($MgSO_4 \cdot H_2O$) . . . 68
5.3.2. Production of Epsom Salt . . . 69
5.3.3. Production of Anhydrous Magnesium Sulfate . . . 69
5.4. Quality Specifications and Analysis . . . 69
5.5. Transportation and Storage . . . 70
5.6. Uses . . . 70
5.7. Economic Aspects . . . 71
5.8. Toxicology and Occupational Health . . . 71
6. Organomagnesium Compounds . . . 71
6.1. Properties . . . 71
6.2. Production . . . 72
6.2.1. Grignard Reagents . . . 72
6.2.2. Diorganylmagnesium Compounds . . . 73
6.3. Uses . . . 74
References . . . 74

1. Introduction

Magnesium occurs in divalent form in all of its compounds. The bromide, iodide, sulfate, and nitrate salts are water soluble. Some of the water-soluble salts are highly hygroscopic and form crystals with a high water content. Magnesium fluoride, oxide, hydroxide, phosphate, and carbonate are sparingly soluble or insoluble in water.

The water-soluble magnesium salts tend to undergo hydrolysis and form basic salts due to the weakly basic character of magnesium hydroxide. This is responsible for the highly corrosive action of magnesium salt solutions; dehydration of magnesium salts containing water of

© 2012 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim
DOI: 10.1002/14356007.a15_595.pub2

crystallization without hydrolytic decomposition is also difficult to achieve.

The most important magnesium compounds are magnesium oxide, which is used for refractory magnesia bricks and as a heat-storage medium, and magnesium chloride which is electrolyzed to produce magnesium metal. Naturally occurring magnesium carbonate (magnesite) is mainly burned to give sintered magnesia for the production of magnesia bricks. Seawater is an important source of magnesium oxide and magnesium chloride. Calcined dolomite ($CaCO_3 \cdot MgCO_3$) is used as a raw material for the silicothermic production of magnesium metal (→ Magnesium). Basic magnesium carbonate is used for thermal insulation.

Anhydrous magnesium nitrate [*10377-60-3*] is difficult to produce (*mp* of the hexahydrate [*13446-18-9*] 89 °C). Concentrated magnesium nitrate solutions (e.g., from $MgCO_3$ and 60% nitric acid) are used for the production of highly concentrated nitric acid (→ Nitric Acid, Nitrous Acid, and Nitrogen Oxides).

Naturally occurring magnesium silicates include asbestos (→ Asbestos) and talc (→ Talc). Materials for electrical insulation are also based on magnesium silicates (→ Insulation, Electric, Section 3.1.2.2., → Insulation, Electric, Section 3.1.2.3.). A series of refractory ceramics are based on magnesium (→ Refractory Ceramics).

Organomagnesium compounds are also known, the most important of these being the Grignard reagents which play an important role in synthetic organic chemistry.

Other magnesium compounds are dealt with elsewhere in the Encyclopedia: magnesium cyanide, → Cyano Compounds, Inorganic, Section 2.2.; magnesium fluoride, → Fluorine Compounds, Inorganic, Section 4.4.; magnesium hydride, → Hydrides, Section 2.6.; magnesium ammonium phosphate, → Fertilizers, 2. Types, Section 2.4.4.

2. Magnesium Carbonate

Properties

Anhydrous Magnesium Carbonate [13717-00-5], $MgCO_3$, M_r 84.33, occurs naturally as *magnesite* which is the starting material for the production of magnesia, MgO (see Section 4.3). Magnesium carbonate is also synthesized by reacting $Mg(OH)_2$ with carbon dioxide at high pressure [1]. Magnesium carbonate begins to decompose at ca. 400 °C, the reaction proceeds rapidly above 550 °C. The compound forms colorless or white to gray, trigonal crystals: lattice constant $a = 0.585$ nm, $\alpha = 103°20'$, Z 4, cleavage parallel to (1011).

Density	2.96 g/cm^3
Mohs hardness	4–4.5
Refractive index	n_0 1.700, n 1.509
Enthalpy of formation (298 K)	− 1096 kJ/mol
Dissociation energy (298 K)	+ 1190 kJ/kg
Solubility in cold water	0.0106 wt%

Barringtonite [5145-48-2], $MgCO_3 \cdot 2\,H_2O$, M_r 120.35, forms colorless triclinic crystals, lattice constants $a = 0.9115$ nm, $b = 0.6202$ nm, $c = 0.6092$ nm; $\alpha = 94.00°$, $\beta = 95.53°$, $\gamma = 108.87°$; $Z = 4$; refractive index 1.458, 1.473, 1.501; density 2.83 g/cm^3.

The mineral *nesquehonite* [*14457-83-1*], $MgCO_3 \cdot 3\,H_2O$, M_r 138.37, occurs as colorless to white crystals. This carbonate can also be synthesized by carbonation of $Mg(OH)_2$ at atmospheric pressure. Density 1.84 g/cm^3, *mp* 165 °C (decomp.), solubility in cold water 0.129 wt%.

Lansfordite [61042-72-6], $MgCO_3 \cdot 5\,H_2O$, M_r 174.40, forms white monoclinic crystals, density 1.73 g/cm^3.

The natural mineral *hydromagnesite* [*12072-90-1*], $4\,MgCO_3 \cdot Mg(OH)_2 \cdot 4\,H_2O$, M_r 467.67, forms white monoclinic crystals, density 2.25 g/cm^3 [2]. The synthetic product is known as magnesia alba.

Other $MgCO_3$-containing minerals include *huntite* [*19569-21-2*], $CaCO_3 \cdot 3\,MgCO_3$, M_r 353.05 [2], and *dolomite* [*16389-88-1*] $CaCO_3 \cdot MgCO_3$, M_r 184.41. Dolomite is a rock-forming mineral and is much more common than magnesite. It is burned to form sintered dolomite (for refractory bricks) and light-burned dolomite (used as a fluxing agent).

Uses. The main use of magnesite is the production of magnesia (Section 4.3). Small quantities are used as a filler. Plans have been proposed for producing magnesium metal from naturally occurring (cheap) magnesite [3]. The synthetic carbonates are used as fillers for paper, plastics, and rubber and as thickening agents for printing inks. Pharmaceutical-grade magnesium

carbonate (magnesia alba) is also used in toothpaste and as an antacid. High-purity magnesium carbonate is employed in cosmetics.

3. Magnesium Chloride

Magnesium chloride [*7786-30-3*], $MgCl_2$, is the most common naturally occurring, water-soluble magnesium compound. It is a constituent of many salt lakes and natural brines [4, 5]; sea salt contains 17 wt% $MgCl_2$. Mineral salt deposits contain the important minerals carnallite [*1318-27-0*] $KCl \cdot MgCl_2 \cdot 6\ H_2O$, bischofite [*7791-18-6*] $MgCl_2 \cdot 6\ H_2O$, and occasionally the double salt tachhydrite in which the magnesium chloride is combined with calcium chloride ($CaCl_2 \cdot 2\ MgCl_2 \cdot 12\ H_2O$). Tachhydrite is found in salt deposits in Brazil and the Congo region.

Magnesium chloride is obtained from mineral salt deposits by processing the spent liquors remaining after extraction of potassium chloride and also by direct solution mining; it is also recovered from salt lakes and seawater.

3.1. Properties

Anhydrous Magnesium Chloride. Anhydrous magnesium chloride forms hexagonal crystals, its physical properties are as follows:

M_r	95.23	
mp	713 °C	
Refractive index (*n*)	1.675, 1.59	
d_4^{20}	2.41 g/cm^3	
Specific heat (c_p) between 0 and 713 °C (*mp*): $c_p = 0.78011 + 0.20607\times10^{-3}\times(t-20) - 0.10853\ 10^{-6}\times(t-20)^2$ kJ kg^{-1} K^{-1} (t in °C)		
Specific electrical conductivity at		
700 °C	1.00 Ω^{-1} cm^{-1}	
750 °C	1.09 Ω^{-1} cm^{-1}	
800 °C	1.17 Ω^{-1} cm^{-1}	
Solubility in water at		
20 °C	35.3 wt%	54.6 g/100 g
40 °C	36.5 wt%	57.5 g/100 g
60 °C	37.9 wt%	61.0 g/100 g
80 °C	39.8 wt%	66.1 g/100 g
100 °C	42.2 wt%	73.0 g/100 g
Solubility in methanol at		
0 °C	15.5 g/100 g	
60 °C	20.4 g/100 g	
Solubility in ethanol at		
0 °C	3.61 g/100 g	
60 °C	15.89 g/100 g	

Table 1. Transformation temperatures and corresponding solubilities of magnesium chloride hydrates

Hydrate transformation	Temperature, °C	Solubility, g $MgCl_2$/100 g H_2O
$MgCl_2$ solution→$MgCl_2 \cdot 12\ H_2O$	− 33.6	27.2
$MgCl_2 \cdot 12\ H_2O$→$MgCl_2 \cdot 8\ H_2O$	− 17.4	47.6
$MgCl_2 \cdot 8\ H_2O$→$MgCl_2 \cdot 6\ H_2O$	− 3.4	52.3
$MgCl_2 \cdot 6\ H_2O$→$MgCl_2 \cdot 4\ H_2O$	116.8	85.6
$MgCl_2 \cdot 4\ H_2O$→$MgCl_2 \cdot 2\ H_2O$	181.5	125.7

Properties of the Hydrates. Magnesium chloride forms five hydrates whose transformation temperatures and solubilities are given in Table 1 (see also Fig. 1). The most important is the hexahydrate, $MgCl_2 \cdot 6\ H_2O$, which forms monoclinic crystals and is present in mineral salt deposits:

M_r	203.31
mp	117 °C
Heat of fusion	34.3 kJ/mol
Density	1.57 g/cm^3
Refractive index (*n*)	1.495, 1.507, 1.528
Specific heat (c_p) at 48.6 °C	1.415 kJ kg^{-1} K^{-1},
100 °C	1.537 kJ kg^{-1} K^{-1}

The penta- and hexahydrates can be dehydrated to the tetrahydrate stage almost without decomposition. On dehydration to the dihydrate, however, hydrolytic decomposition starts. Dehy-

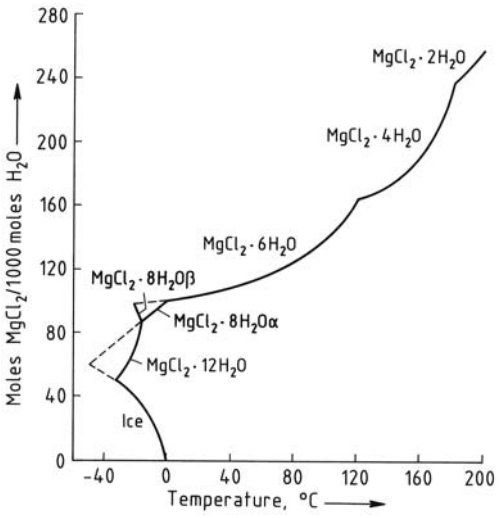


Figure 1. State diagram of the system $MgCl_2 \cdot n\ H_2O$

dration of the dihydrate at 240 °C leads to extensive decomposition forming magnesium oxychlorides or oxide. To obtain anhydrous $MgCl_2$, the dihydrate must be dehydrated in an atmosphere of hydrogen chloride.

Chemical Properties. Magnesium chloride can be decomposed by oxygen to form magnesium oxide and chlorine [6] – the converse of the reaction between magnesium oxide and chlorine to form $MgCl_2$. With ammonia, $MgCl_2$ forms ammines that are analogous to the hydrates: $MgCl_2 \cdot NH_3$, $MgCl_2 \cdot 2\ NH_3$, and $MgCl_2 \cdot 6\ NH_3$. Corresponding addition compounds are formed with alcohols and with other organic compounds such as ethers, acetic acid, amines, ethyl acetate, pyridine, hexamethylenetetramine, and dioxane (e.g., $MgCl_2 \cdot 6\ CH_3COOH$).

A series of basic chlorides (magnesium oxychlorides) are formed with magnesium oxide and are of commercial importance as Sorel cement (see 3.8). The following oxychlorides are known: $MgCl_2 \cdot MgO$, $MgCl_2 \cdot MgO \cdot H_2O$, $2\ MgCl_2 \cdot MgO \cdot H_2O$, $MgCl_2 \cdot 3\ MgO \cdot 11\ H_2O$, and $MgCl_2 \cdot 3\ MgO \cdot 7\ H_2O$. The compounds $MgCl_2 \cdot MgO$ and $2\ MgCl_2 \cdot MgO \cdot H_2O$ are formed on dehydration of $MgCl_2 \cdot H_2O$ at 280–495 °C. Aqueous $MgCl_2$ solutions are weakly acidic due to hydrolysis. The electrolysis of aqueous $MgCl_2$ solutions produces magnesium hydroxide and chlorine, whereas electrolysis of anhydrous $MgCl_2$ yields magnesium metal and chlorine.

3.2. Raw Materials

Of the three main sources of raw materials for magnesium chloride production, the most important is *seawater*. Magnesium can be directly precipitated from seawater as magnesium hydroxide and converted to magnesium chloride by treatment with chlorine. The residues (bitterns) obtained after production of drinking water or sea salt from seawater can also be used. A second large source is the *spent liquor from the potash industry* arising from the processing of hard salts and carnallite-bearing raw materials. A third source is provided by the *salt lakes* in many parts of the world, the Dead Sea and the Great Salt Lake being particularly important.

3.3. Production

3.3.1. Precipitation of Magnesium Hydroxide from Seawater

If seawater cannot be preconcentrated by solar energy, magnesium chloride salts or solutions can only be produced on a large scale by precipitation as the hydroxide, followed by conversion to the chloride with hydrogen chloride.

The Dow Chemical process is described elsewhere (→ Magnesium). This process has been used to manufacture magnesium chloride from seawater in large quantities since 1941 in Freeport (Texas). It is an intermediate step in the production of magnesium metal.

3.3.2. Recovery from Spent Liquors from the Potash Industry

The spent liquors obtained after recovery of potassium chloride from carnallite-containing raw salts (→ Potassium Compounds) typically contain (in grams per liter): $MgCl_2$ 293, $MgSO_4$ 48, KCl 39, NaCl 26, and H_2O 890. They are used as a raw material for the recovery of magnesium chloride. For the Norsk Hydro process, see → Magnesium.

Production of Concentrated $MgCl_2$ Solutions by Kali und Salz AG (Germany). The spent liquor contains ca. 280–300 g/L $MgCl_2$ and is concentrated in a two-stage vacuum evaporator to ca. 460 g/L $MgCl_2$. (Vapor pressures of $MgCl_2$ solutions as a function of concentration and temperature are shown in Fig. 2).

The solution in the first evaporator stage is heated by the vapor from the second stage to a steady temperature of ca. 55 °C. The second-stage liquor is maintained at 95 °C by heating with steam at 0.1 MPa and the hot, cloudy solution is clarified by removing the salts [carnallite, NaCl, $MgSO_4 \cdot 5/4\ H_2O$, and impurities such as $Fe(OH)_3$ and clay] by means of a rotary filter and centrifuge. On cooling carnallite crystallizes from the solution. The crystallized salts and sludge are sent to the potassium chloride plant where carnallite is decomposed together with carnallite containing crude potash salt. The carnallite is mixed with water or a low-$MgCl_2$ brine to produce a spent liquor containing 280–300 g/L

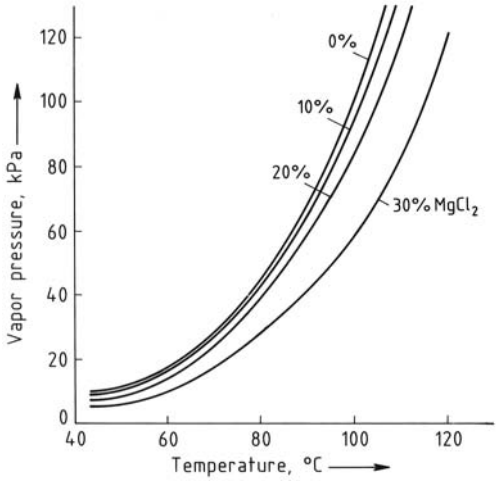


Figure 2. Vapor pressure of $MgCl_2$ solutions as a function of temperature

of $MgCl_2$ and solid potassium chloride contaminated with sodium chloride and $MgSO_4 \cdot 5/4\ H_2O$ (→ Potassium Compounds). The cooled $MgCl_2$ solution can be purified by reheating and removing bromine with chlorine. In this case the debrominated solution is neutralized with lime, partly precipitating the sulfate ions in the form of gypsum. The precipitated sediment is removed. The highly concentrated 32–33% $MgCl_2$ solution is sold as such (see Section 3.5) or can be processed further to obtain magnesium chloride hydrates (see Section 3.3.6).

Production of Concentrated $MgCl_2$ Solution by the VEB Kombinat Kali (former German Democratic Republic). The spent liquor from the manufacture of potassium salts had a $MgCl_2$ content of ca. 260 g/L. The solution was first mixed with synthetic carnallite from the vacuum cooling stage; the $MgCl_2$ from the carnallite dissolves leaving the undissolved potassium chloride behind. The increased $MgCl_2$ content (ca. 320 g/L) of the solution lowers the solubility of the alkali-metal salts causing the precipitation of potassium and sodium chlorides. After removal of these salts, the solution was pumped to the second of the two evaporation stages. The preconcentrated solution from the second evaporation stage was pumped to the first evaporation stage where $MgSO_4 \cdot 5/4\ H_2O$ precipitates together with some langbeinite, carnallite, and sodium chloride. On cooling this solution, (after removal of the salts) in multistage vacuum coolers, synthetic carnallite precipitates. The carnallite was filtered off and recycled to the first step of the process. All the evaporators were of the forced circulation type [7]. The first evaporator stage was operated at 120 °C and heated with steam. The second stage was heated to 80 °C by the vapor from the first stage. The vapors from the second stage were cooled with water.

In a variation of the process, the solution containing 320 g/L $MgCl_2$ was mixed with part of the product solution (containing 460 g/L $MgCl_2$). The increase in the $MgCl_2$ content causes more sodium chloride and carnallite to precipitate. Evaporation of the solution from the first stages then does not precipitate carnallite or langbeinite ($K_2SO_4 \cdot 2\ MgSO_4$). Only synthetic kieserite ($MgSO_4 \cdot 5/4\ H_2O$) and sodium chloride are precipitated and these are removed. The solution produced had a density of 1.340 g/cm^3 and contained 460 g/L $MgCl_2$, 25 g/L $MgSO_4$, 2 g/L KCl, 6 g/L NaCl, 2 g/L $CaSO_4$, and 4 g/L Br^-. If these solutions are used for the production of magnesium metal, they must be debrominated with chlorine (→ Bromine, Chap. 4.). If they are used for production of high-quality sintered magnesia, boron is also removed by boron-selective resins. The boron content decreases from 30 to 5 –10 mg/L [8].

Recovery from Raw Salts Containing Carnallite. Patents have been applied for in which pure carnallite or raw salts containing carnallite are decomposed with C_1– C_4 alcohols, causing the $MgCl_2$ to pass into the alcohol phase [9]. The alcoholic solution is concentrated by evaporation to an alcohol content of 10%, and the hexahydrate is precipitated by addition of water. Alternatively, evaporation is continued until the solution contains 1–2 mol of alcohol per mole of $MgCl_2$; the alcoholate is then thermally decomposed in a hydrogen chloride atmosphere at 400–500 °C.

3.3.3. Evaporation of Seawater and Natural Brines

The direct production of $MgCl_2$ from seawater or brines by evaporation is only economical where the dilute solutions can be preconcentrated by solar evaporation (→ Sodium Chloride).

Recovery from the Dead Sea. Magnesium chloride-containing solutions are produced in large quantities by solar evaporation of brine from the Dead Sea. At the Dead Sea Works, carnallite precipitates in the final stage, and solutions containing 360 g/L $MgCl_2$, 110 g/L $CaCl_2$, 7 g/L NaCl, and 5 g/L KCl are obtained. Owing to the low vapor pressure of water, solar evaporation only occurs at very low atmospheric humidity levels. After two or three months of evaporation, a density of 1365 g/L is achieved and bischofite crystallizes [10]. Large quantities of $MgCl_2$-containing liquors of this type are now used by Dead Sea Periclase solely for the production of magnesium oxide by thermal decomposition.

In a process patented by the Dead Sea Works for the manufacture of magnesium chloride hexahydrate (bischofite), carnallite or a natural mixture of carnallite and sodium chloride is decomposed in its own water of crystallization in closed vessels under pressure at 167.5 °C. The potassium chloride remains together with the sodium chloride as a sediment and is removed. The residual solution contains up to 8.2% potassium chloride and 42.3% magnesium chloride. On cooling to 115 °C by flash evaporation, carnallite is precipitated and is fed back into the circulating system. The residual solution contains 1% potassium chloride and consists essentially of bischofite [11].

Recovery from the Kara–Bogaz Gulf (Turkmenia). Following the isolation of the Kara-Bogaz Gulf from the rest of the Caspian Sea by a dam in 1979, the $MgCl_2$ content increased to 250 g/L. Research was carried out to find a new method of processing this brine [12]. The brines evaporated in summer contain 25–30 wt% $MgCl_2$. When they are cooled in winter, epsomite ($MgSO_4 \cdot 7\ H_2O$) precipitates and is used for producing schönite ($K_2SO_4 \cdot MgSO_4 \cdot 6\ H_2O$). The liquid phase is used for $MgCl_2$ production.

Recovery from Spent Liquors after Extraction of NaCl from Seawater. The spent liquors (sea bitterns) obtained after extraction of salt from seawater are evaporated in crystallizing pans by solar energy to a density of 1290 g/L. After removing the sodium chloride, the solution is cooled to 5 °C, which can be achieved in winter without energy consumption. Magnesium sulfate (epsomite) crystallizes with a few impurities and is removed. The bitterns are cooled to 0 °C; sodium sulfate (Glauber's salt) crystallizes and is removed. The solution is then debrominated with chlorine (→ Bromine, Chap. 4.) to exclude crystallization of magnesium bromide in subsequent processing. After neutralizing with potassium hydroxide, the liquor is heated to 125 °C. On cooling to 100 °C, a crude potassium-containing salt mixture separates that consists of langbeinite ($K_2SO_4 \cdot 2\ MgSO_4$) and kainite ($KCl \cdot MgSO_4 \cdot 3\ H_2O$), with some kieserite ($MgSO_4 \cdot H_2O$) and sodium chloride. On further cooling to 20 °C, carnallite crystallizes and can be treated with water to form a $MgCl_2$-containing solution and solid potassium chloride [13].

Recovery from the Great Salt Lake. For the National Lead Industries process, see → Magnesium. In the process operated by the Great Salt Lake Minerals Chemical Corporation, a 35-wt% $MgCl_2$ solution is first produced by solar evaporation, and this is then subjected to vacuum evaporation until bischofite ($MgCl_2 \cdot 6\ H_2O$) and hydrated $MgSO_4$ are precipitated. Proposed further treatment consists of heating the crystals to 120–150 °C to redissolve the bischofite and allow removal of magnesium sulfate. The bischofite can then be crystallized [14]. Alternatively, the solution can be maintained at 120 °C until the magnesium sulfate content decreases to < 20 g/L as a result of crystallization of $MgSO_4 \cdot 5/4\ H_2O$. The bischofite is then crystallized by vacuum evaporation at 90 °C [15].

Recovery from Naturally Occurring Brines with High $CaCl_2$ Contents. Up to the end of World War II, the Dow Chemical Company used underground brines from Midland (Michigan) and Marysville for bromine production and also for the production of $MgCl_2$. The latter was electrolyzed to obtain magnesium. The alkali metal salts were removed by evaporation and crystallization. In an older process the calcium was then separated together with the magnesium as tachhydrite ($2\ MgCl_2 \cdot CaCl_2 \cdot 12\ H_2O$) which was converted into $MgCl_2 \cdot 6\ H_2O$. In a later process, the solution was reacted with an amount of magnesium hydroxide (produced from magnesite) equivalent to the calcium chloride content. The calcium was then removed as calcium

carbonate by treating the solution with the carbon dioxide from the magnesite calcination, and the $MgCl_2$ was obtained from the solution.

Today, Dow obtains magnesium chloride for magnesium production only from seawater (→ Magnesium). The production of bischofite from Michigan brines has been discontinued.

3.3.4. Solution Mining

A 20-wt% $MgCl_2$ solution is produced in large quantities in the Netherlands from a carnallite salt deposit. It is used to produce concentrated $MgCl_2$ solutions and is converted to magnesium hydroxide (see Section 4.3.3) [16].

Solution Mining of Carnallite. Carnallite ($KCl \cdot MgCl \cdot 6\ H_2O$) deposits contain large amounts of magnesium sulfate as well as $MgCl_2$. Evaporation of solutions with high magnesium sulfate contents leads to precipitation of double salts containing sodium chloride and magnesium sulfate. In the processing of these solutions, most of the magnesium sulfate is precipitated as langbeinite ($K_2SO_4 \cdot 2\ MgSO_4$) [17], or as schönite ($K_2SO_4 \cdot MgSO_4 \cdot 6\ H_2O$) by precipitation with methanol [18]. In a process developed by the Mines de Potasse d'Alsace, a solution containing 320 g/L $MgCl_2$ can be obtained together with solid potassium chloride by solution mining. The solution used to extract the carnallite from the deposit contains a flotation agent for potassium chloride (C_{12}–C_{20} fatty amines or their salts) and a gas-producing agent such as hydrogen peroxide. The potassium chloride floats and can be pumped out along with the $MgCl_2$ solution [19]. This production method appears promising, but it is not known whether large-scale development is planned.

3.3.5. Other Methods

Magnesium chloride can be produced from dolomite in combination with soda by the Solvay process. In the early part of World War II at a plant belonging to the Diamond Alkali Corporation in Painesville, Ohio, the dolomite was calcined in rotary furnaces, hydrated, and used to regenerate ammonia from the ammonium chloride solution produced in soda manufacture. Only the calcium hydroxide took part in the reaction. On treatment of the remaining solution with carbon dioxide, a magnesium chloride solution was obtained:

$$Mg(OH)_2 + CaCl_2 + CO_2 \rightarrow MgCl_2 + CaCO_3 + H_2O$$

This process is, however, no longer used.

In the Mathison Alkali Works, the calcium chloride solution from the soda process was treated with calcined dolomite while carbon dioxide was introduced. This process also produced a $MgCl_2$ solution but has been discontinued.

Calcined magnesite or magnesium hydroxide can also be used in the soda process to produce a $MgCl_2$ solution.

Processes are described in which magnesium chloride is produced from natural crude magnesite by leaching with hydrochloric acid, as well as the use of phosgene and crude hydrochloric acid with 27.5% HCl and 1.2% HF to treat magnesite and periclase [20–22].

3.3.6. Production of Magnesium Chloride Hydrates

Production of $MgCl_2 \cdot 6\ H_2O$ (Bischofite). In a process formerly used by Kali und Salz AG, end bittern from potash production was treated in a single-stage vacuum evaporator to produce a ca. 42-wt% $MgCl_2$ solution, and was further evaporated in pans with steam at 1.1 MPa. After neutralization with lime and removal of calcium sulfate, synthetic kieserite, and other impurities, the melt was fed onto rotating drum coolers or cooling bands to form a slab, which was broken up and marketed (see Section 3.5). Alternatively, after the initial single-stage evaporation, the solution can be prilled in a prilling tower or spray dried in a fluidized bed [23].

A very pure $MgCl_2$ hydrate with 4–6 molecules of water can be prepared in 99.9% purity according to a patent of the VEB Kombinat Kali [24]. Carnallite is dehydrated at 180 –200 °C and then cooled. The cooled salt is then treated with acetone, in which the $MgCl_2$ dissolves. After removal of the undissolved salt, the hydrate is precipitated from the acetone solution by adding the correct amount of water [24].

In a patent of the Mines de Potasse d'Alsace, $MgCl_2$ is precipitated from solution by addition

of 10–25% 1,4-dioxane at room temperature. At lower temperatures the yield of $MgCl_2$ is increased. The $MgCl_2$– dioxane complex separates as a voluminous precipitate that is easy to filter or centrifuge. It can be purified by washing with a little dioxane. The dioxane can be distilled off from the filtrate as an 80% azeotrope and can be reused directly. The $MgCl_2$ hydrate can be obtained by washing with water. During drying the remaining dioxane (ca. 20–25% of the total) is evaporated off and can also be reused [25].

Production of $MgCl_2 \cdot (2–3)H_2O$. In a process formerly used by Kali und Salz AG, a ca. 33-wt% $MgCl_2$ solution was heated to 110–130 °C and sprayed into a fluidized bed of magnesium chloride granules at 130 °C. The heating gas was at 400–600 °C. The average residence time of the granules is ca. 30 min. Fines and pulverized oversize granules were recirculated to the fluidized bed. The final product had a water content of ca. 30% [23].

3.3.7. Production of Anhydrous Magnesium Chloride (See also → Magnesium, Section 4.1.1.)

The Nalco Chemical Company at its plant in Freeport, Texas, produces tetramethyl- and tetraethyllead. The Grignard reagents needed as intermediates for the process require a very pure source of magnesium. This is produced by the electrolysis of pure, anhydrous magnesium chloride, for which the following process has been developed.

Ethylene glycol is added with stirring to a 30–35-wt% $MgCl_2$ solution, the quantity added being sufficient to produce a 10–20-wt% $MgCl_2$ solution. All the water, including water of crystallization, is then removed by vacuum distillation. The distillate contains < 50 ppm ethylene glycol, and the residual solution contains < 25 ppm water together with all the other salts. Gaseous ammonia is then bubbled through the solution under slight pressure. The mixture is cooled to remove the evolved heat; in practice the temperature is allowed to rise to ca. 70 °C. When the glycol is saturated with ammonia, the hexaammine compound $MgCl_2 \cdot 6\ NH_3$ is formed. Towards the end of the reaction, the mixture is cooled to 15–30 °C, the $MgCl_2 \cdot 6\ NH_3$ precipitates and is filtered off. Any adhering solution is removed by washing with methanol. The ammonia and methanol are then removed in two stages: the methanol and some ammonia are first driven off at 74 °C. The rest of the ammonia is then removed in a calcination furnace at 400 °C [26].

As with the dehydration of carnallite, ammonium carnallite can be dehydrated almost without decomposition at 90–180 °C. It can then be decomposed at 180–400 °C to give ammonium chloride and anhydrous $MgCl_2$ [27, 28].

In the process developed by Esso Research, a 20–50-wt% solution of magnesium chloride reacts with ammonium chloride in a three-stage fluidized bed. Anhydrous magnesium chloride is produced at 350–550 °C [29].

In another process [30], ammonium carnallite containing n NH_3 or n H_2O is decomposed at 800 °C. A product with a low magnesium oxide content is obtained. If the starting material contains water of crystallization, some is removed in a first stage, and further dehydration takes place after adding ammonia. After the dehydration, the $MgCl_2 \cdot 6\ NH_3$ is thermally decomposed [31].

A process for the dehydration of magnesium chloride hydrates in a hydrogen chloride atmosphere is described in → Magnesium, Section 4.1.1.2.

3.3.8. Purification of Magnesium Chloride Solutions

Calcium ions can be removed from magnesium chloride solutions by adding the stoichiometric amount of sulfate ions [32]. Conversely, sulfate ions can be removed by adding calcium chloride solutions. Calcium and other metal ions can be extracted with dialkylphosphoric acid in a water-immiscible solvent. This process is used in the production of pure sintered magnesia from solutions of impure magnesites in hydrochloric acid [33]. These solutions can be treated with substances such as MgO, CaO, or $Mg(OH)_2$ to increase the pH, while simultaneously purging with air or oxygen at 80–90 °C. This removes impurities such as iron, aluminum, and manganese as hydrated oxides or hydroxides. This process is also used in the manufacture of pure sintered magnesia [34].

3.3.9. Materials of Construction for the Evaporation of Magnesium Chloride Solutions

At temperatures up to 110 °C, corrosion is usually prevented by the use of coatings, rubber linings, or plastic reinforced with glass fibers. Heater tubes and distillation trays are usually made of aluminum bronze ($AlBz_{5-7}$) as used in the potash industry, with a composition of 4 –8% Al, 89–96% Cu, and small amounts of Fe, Ni, Mn, As, Si, or Zn. Pumps are made of NiResist D 2.

Above 110 °C, and especially in the presence of chlorine and bromine (during debromination), glass and Teflon are suitable, with titanium being used for preheaters and pipework. According to [35] brick-lined steel vessels are used for evaporating $MgCl_2$ solutions and steam-heated tubes are made of Inconel alloy or stainless steel 347. Monel metal is also used for > 50% $MgCl_2$ solutions.

3.4. Environmental and Legal Aspects

Magnesium chloride is a constituent of seawater. In the EEC Regulations (May 4th 1976) concerning disposal of hazardous substances in lakes and rivers, this substance is not mentioned. However, if large quantities are discharged into lakes or rivers, the concentration can be high enough to injure aquatic organisms and immediate dilution must be carried out. The WHO international standard for drinking water for magnesium is 30 mg/L Mg, or 125 mg/L if the sulfate concentration is < 250 mg/L and the chloride concentration is < 200 mg/L. In the EEC regulation 80/778 EWG the limit for drinking water is 50 mg/L Mg and 25 mg/L Cl^-. In the dehydration of magnesium chloride hydrates, hydrogen chloride occurs in the waste gases due to partial decomposition to magnesium oxide and the concentration must be kept below the permitted limit by use of gas purification equipment. The German TA Luft Regulations specify a maximum of 30 mg/m^3 HCl in air. Limits in Australia are 200–400, in the United Kingdom 150, and in Japan 80–700 mg/m^3.

3.5. Quality Specifications and Analysis

Quality Specifications. Magnesium chloride is available commercially in the form of solutions or solid hydrates. Solutions usually have a $MgCl_2$ content of 32–33 wt%. A typical solution with a $MgCl_2$ content of 32–33 wt% has the following approximate composition (in grams per liter);

$MgCl_2$	430–440
$MgSO_4$	12–25
KCl	2–3
NaCl	4–6
H_2O	ca. 870

For many applications, however, solutions produced by decomposition of carnallite (see Section 3.3.2) are used directly; due to transport costs this only applies when the user is close to the supplier.

The most important hydrate sold commercially is the hexahydrate (bischofite) which contains 43–47-wt% $MgCl_2$, ca. 51.5 wt% water, ca. 2 wt % alkali metal chlorides, and, depending on its origin, some $MgSO_4$, $CaSO_4$, or $CaCl_2$. It is supplied in the form of broken pieces or flakes, or as blocks produced by melting and pouring into drums. Another hydrate in granular form contains ca. 2–3 mol of water per mole of $MgCl_2$ and corresponds to 67 wt% $MgCl_2$. Anhydrous magnesium chloride is mostly used directly by the producer for electrolytic magnesium production (→ Magnesium).

Analysis. For quality control the $MgCl_2$ solutions and the solid hydrates must be analyzed for the following ions: Mg^{2+}, Ca^{2+}, K^+, Na^+, Cl^-, SO_4^{2-}, and Br^-. In special cases, determination of boron, iron, and other ions may be required.

Magnesium and calcium are titrated complexometrically with ethylenediaminetetraacetic acid (EDTA) [36, 37]. Potassium and sodium are determined by flame photometry [38], or by ICP [39]. For the determination of sulfate ions, a titration with barium perchlorate solution in a 2-propanol–water mixture is carried out with thorin as the indicator [40] or it is analyzed by ICP. Bromide is determined gravimetrically by the method of D'Ans and Höfer [41].

The water content of the hydrates is determined by heating to 450–600 °C or by the lead oxide method [42]. Boron can be determined photometrically with azomethine-H or dianthrimide-1-amine [43, 44] and iron with bathophe-

nanthroline (4,7-diphenyl-1,10-phenanthroline) [45]. Iron can also be determined by atomic absorption spectrophotometry [46] or by ICP.

3.6. Transportation and Storage

Magnesium chloride solutions can be transported in tank wagons. Owing to their corrosive properties, they should be stored in plastic containers reinforced with glass fiber, but iron vessels can also be used. When a 32–33-wt% $MgCl_2$ solution is cooled, $MgCl_2$ hydrates precipitate, gradually forming a sludge. At–33.6 °C the residual solution containing 21.4 wt% $MgCl_2$ freezes as a eutectic mixture. The hexahydrate is transported in polyethylene sacks or big bags capable of holding > 50 kg. It is also supplied in the form of broken pieces or solid blocks produced by pouring molten salt in single-journey steel drums. Sealed vessels can be stored almost indefinitely, but, after opening, the material must be used immediately because of its hygroscopicity. All containers must be stored under dry conditions.

3.7. Uses

Anhydrous magnesium chloride is used chiefly for the electrolytic production of magnesium metal (→ Magnesium). Large quantities of magnesium chloride solutions are used to produce anhydrous magnesium chloride for this purpose.

An important area of application for the hexahydrate and the hydrate containing 67% $MgCl_2$ is in the production of Sorel cement. This is made by mixing magnesium oxide with $MgCl_2$ solutions, and is mainly used as a binder for industrial floorings on account of its high elasticity and bending tensile strength properties [47]. This binder is also used in the manufacture of abrasive materials and grindstones, and for cementing rock faces during coal mining operations [48]. In combination with wood chippings, it can be used to make lightweight building slabs. Sorel cement mixtures are also used in water- and fire-resistant building materials [49]. They also serve as a binder for granulated fertilizers and in the production of salt licks for animals, where the magnesium also acts as a feed additive, preventing diseases such as grass tetany in cattle that are caused by magnesium deficiency [50].

Another application for $MgCl_2$ solutions and the hexahydrate is the manufacture of sintered magnesia and very pure $Mg(OH)_2$ (see Section 4.3.5). A high proportion of the solutions produced in Israel and the Netherlands are used for this purpose. Magnesium chloride hexahydrate and potassium chloride are used to produce artificial carnallite, which is used as a covering flux in the melting of light metal alloys. In the Quentin process used in the sugar industry, the alkali metal ions in the dilute sugar syrup are replaced by magnesium ions by treating it with an anion-exchange resin, thus reducing sugar losses in the molasses by improving sugar crystallization [51]. The resin is regenerated with $MgCl_2$ solution. In the mineral oil industry, $MgCl_2$ solutions are used as drilling fluids in rocks with a high salt content. The low freezing point of the 21% $MgCl_2$ solution (< -30 °C) is utilized to prevent wet bulk materials from freezing to a solid mass. It is used as a de-icing agent in winter, as a ballast liquid for the tires of construction machinery and agricultural vehicles, for fire-extinguishing foams at low temperatures, and as brine for refrigeration. It is used in road surfaces combined with minerals and asphalt as a antifreeze. It is also used as a heat storage medium in solar heating installations, as an additive in the steel industry, as a coagulant in wastewater treatment, and in the treatment of combustion gases ($MgCl_2$ solution absorbs HCl and improves the removal of SO_2 by lime). In the plastics industry it is used in the manufacture of polymerization catalysts, e.g., in combination with $TiCl_4$ in Ziegler–Natta catalysts for the polymerization of C_2–C_{12} olefins. Addition of $MgCl_2$ also reduces losses of ammonia that is used as a fertilizer [52].

$MgCl_2$ is used in the production of moisture-proof salts. It is also used in the pharmaceutical and cosmetic industries, in food- and feedstuffs, and in bath salts. In addition it is used in the production of other Mg salts such as magnesium-chlorate.

3.8. Economic Aspects

The total worldwide production of $MgCl_2$ in 1996 was 1.5×10^6 t. In the world export mar-

kets ca. 300 000 t $MgCl_2$ was handled: ca. 200 000 t in the form of 33% $MgCl_2$ solutions, and 100 000 t in the form of the hexahydrate. About half of the $MgCl_2$ solution was used for production of magnesium metal. In addition, approximately 180 000 t $MgCl_2$ from solution mining brines and 150 000 t $MgCl_2$ as 33% solution were converted to magnesium oxide; these quantities were reported as material produced but not marketed. A further ca. 700 000 t magnesium chloride was not recorded in any statistics because it was used solely as an intermediate for the production of magnesium by electrolysis.

The most important countries for the production of $MgCl_2$ solutions and the hexahydrate are given below (excluding magnesium metal producers):

Germany (Kali und Salz AG)	byproduct solutions from the potash industry
Israel	Dead Sea brines
Netherlands	solution mining brines
Japan	sea bitterns from salt production
France	sea bitterns
United States	Great Salt Lake brines
Italy	sea bitterns
China	brines from various salt lakes
Australia	from Kunwarara magnesit deposit

Two thirds of the material appearing on the market originates from Germany.

3.9. Toxicology and Occupational Health

The handling of $MgCl_2$ hydrates, anhydrous $MgCl_2$, or $MgCl_2$ solutions on a laboratory scale does not present any personal hazards. German chemical regulations do not require any special marking. The LD_{50} (rat, oral) for $MgCl_2 \cdot 6\ H_2O$ is 8100 mg/kg. Eye protection and rubber gloves should be worn for handling anhydrous $MgCl_2$, $MgCl_2$ hydrates, or solutions. Prolonged contact with concentrated (ca. 35 wt%) solutions or with $MgCl_2$ dust can cause skin irritation, and possibly allergic skin reactions. In the event of skin problems, the affected area should be washed with copious soap and water, followed by treatment with a fat-containing skin cream. In case of contact with the eye, the eye should be washed with water and medical assistance should be obtained.

4. Magnesium Oxide and Hydroxide

Magnesium Oxide [*1309-48-4*], MgO (magnesia, periclase), seldom occurs as a natural mineral; it is found in contact metamorphic limestone and dolomite, in volcanic ejecta, and in serpentine rocks. It does not form rocks or salt deposits because it is converted to magnesium hydroxide by the water vapor in the atmosphere. The primary sources of industrially produced magnesia are natural magnesite ($MgCO_3$), seawater, and natural and synthetic brines. Magnesium hydroxide is formed as an intermediate in the production of magnesia from seawater and brines. Magnesia is used in technical applications on account of its high melting point, chemical resistance, high thermal conductivity, low electrical conductivity, and biological activity.

Technical-grade magnesia is mainly sintered to form *sintered magnesia* (also known as magnesia sinter, dead-burned magnesia, or sintered magnesite) which is used primarily as a refractory material and in the steel industry. Large quantities of calcined (decarbonated) material known as *caustic magnesia* or caustic-calcined magnesia, are produced for use in agriculture and the building industry. *Fused magnesia* is normally produced by melting caustic magnesia.

Magnesium Hydroxide [*1309-42-8*], $Mg(OH)_2$ (brucite) is produced by precipitation of seawater and brines with calcium hydroxide (Section 4.3.2) or by precipitation of soluble magnesium salts. Natural deposits are rare and are not currently being mined.

4.2. Properties

Magnesium Oxide [*1309-48-4*], MgO (magnesia, periclase [55]), M_r 40.31, is stable in an oxidizing atmosphere up to ca. 2300 °C and up to ca. 1700 °C in a reducing atmosphere.

Crystal Structure. Magnesia forms colorless, green, or brown, cubic hexoctahedral crystals, space group O_h^5–*Fm* 3*m*, ionic radius of

Mg^{2+} 0.078 nm, $a = 0.421$ nm, $Z = 4$, cleavage parallel to (100).

Physical Properties

mp	ca. 2800 °C
Mohs hardness	5.5–6
Density	3.58 g/cm^3
Refractive index at 589 nm	1.736
Enthalpy of formation (298 K)	− 14 900 kJ/kg
Specific heat c_p	
20– 200 °C	1.0 kJ kg^{-1} K^{-1}
20–1000 °C	1.2 kJ kg^{-1} K^{-1}
Thermal conductivity	
pure crystals	
300 °C	22 W m^{-1} K^{-1}
1000 °C	7 W m^{-1} K^{-1}
technical-grade	
300 °C	12 W m^{-1} K^{-1}
1000 °C	5 W m^{-1} K^{-1}
Linear thermal expansion coefficient	
room temperature	6.7×10^{-6} K^{-1}
1000 °C	14×10^{-6} K^{-1}
1800 °C	16×10^{-6} K^{-1}
Modulus of elasticity (sintered magnesia, bulk density 3.51 g/cm^3)	
25 °C	3×10^4 MPa
1000 °C	2.5×10^4 MPa
Dielectric constant (30 Hz–1 MHz)	11
Electrical resistivity	
room temperature	1×10^{14} Ω · cm
500 °C	1×10^{11} Ω · cm
1000 °C	1×10^{7} Ω · cm
1300 °C	1×10^{5} Ω · cm

Magnesium Hydroxide [*1309-42-8*], $Mg(OH)_2$ (brucite), M_r 58.33, starts to decompose at 380 °C.

Crystal Structure. Magnesium hydroxide forms ditrigonal scalenohedral crystals with a typical layer structure, space group D^3_{3d}–$P\bar{3}m1$, $a = 0.313$ nm, $c = 0.474$ nm, $Z = 1$, cleavage parallel to (0001).

Physical Properties

Mohs hardness	2.5
Density	2.38 g/cm^3
Refractive index	*n* 1.566, *n* 1.581
Enthalpy of formation (298 K)	−15 800 kJ/kg
Dissociation energy (298 K)	+ 570 kJ/kg
Hydration energy	−386 kJ/kg
Solubility in cold water	0.0009 wt%

Toxicity. Magnesium oxide and hydroxide are not toxic. Precautions for handling dust must be observed when products with a small particle size are used.

4.3. Production of Magnesium Oxide

Caustic and sintered magnesia are generally obtained from magnesium carbonate or magnesium hydroxide. The production of the latter starting materials is described in Sections 4.3.1, 4.3.2, 4.3.3, 4.3.4. The burning and sintering processes are dealt with in Section 4.3.5. The production, properties, and uses of fused magnesia are described in Section 4.6.

4.3.1. Production from Magnesite [56]

The production of magnesium oxide from magnesite began over 100 years ago in Austria, after the development of processes for the manufacture of steel from high-phosphorus pig iron in crucibles lined with sintered dolomite ($CaO \cdot MgO$).

Occurrence of Magnesite. Magnesia is produced from macro- and cryptocrystalline magnesite. Metamorphic *macrocrystalline magnesite* occurs in two Alpine mountain chains in varying degrees of purity. The rocks mainly consist of the isomorphous carbonate mineral (Mg, Fe)CO_3 (breunnerite), whose calcination results in the inclusion of 1–10% Fe_2O_3 in the periclase lattice. In addition, they contain small amounts of dolomite ($CaCO_3 \cdot MgCO_3$), calcite ($CaCO_3$), phyllosilicates (e.g., talc, mica, chlorites, or clay minerals), graphitic particles, and sometimes quartz. Similar magnesite deposits are exploited in the former Czechoslovakia, in the Pyrenees, in Ural (Satka), South Manchuria (Liaoning), East Brazil, and Canada, both by open-cast and underground mining. The macrocrystalline rocks are always subjected to some form of processing before calcination.

In contrast to the macrocrystalline magnesites, which form extensive or lenticular deposits, *cryptocrystalline magnesite* deposits are irregular and often veined. The rocks are mostly snow white, even if they are not composed of high-purity $MgCO_3$. They are regarded as being formed by the alteration of ultrabasic rocks (e.g., serpentines) through the action of water containing carbon dioxide. Deposits of this type occur in former Yugoslavia, Greece (Chalkidike and Euboa), Turkey, India, Nepal, and South Africa. They contain chalcedony (finely divided

SiO_2), various magnesium silicates, and dolomite as accompanying minerals. The deposits are exploited mainly by open-cast mining, but also by underground mining.

Sedimentary deposits of cryptocrystalline magnesite have been identified in the last 20 years. They are often associated with deposits of hydromagnesite and huntite. The sedimentary deposits are formed in semi-arid basins associated with strongly weathered ultrabasic rocks. The material washed into the basin dissolved and then recrystallized on the basin rim, mostly in the form of porous concretions. A more recent formation of this type has been found in Southwest Turkey [57], while an older deposit has been reported in North Greece [58, 59]. A large deposit of this type was discovered in Australia (Kunwarara) in 1985. The Queensland magnesia project has been planned for the production of sintered and fused magnesia [60, 61].

Magnesite Beneficiation. The insufficient purity of the magnesite and the stringent specifications for the uniformity of sintered magnesia mean that beneficiation of the magnesite is usually necessary. The steps used in the beneficiation process depend on the ore being processed and the required carbonate purity.

The highest quality magnesites, particularly those for refractory applications, are needed for a magnesia product with a high MgO content, a $CaO : SiO_2$ mass ratio of 2–3, and low contents of Fe_2O_3 and Al_2O_3. The presence of accompanying, low-melting minerals can adversely affect the properties of the sintered magnesia (cf. Table 2).

Preliminary sorting of the ore can be achieved by *selective mining*. The ore is then subjected to *selective coarse grinding* in impact crushers. The tailings can be separated from the ore by taking advantage of the difference in hardness; the ore is selectively crushed and removed by screening. Crushing is usually performed in several stages. The final particle size for calcination in shaft kilns is 20–200 mm, while a particle size of 2–60 mm is required for the more commonly used rotary kilns.

The difference in color between the cryptocrystalline magnesite ore and the tailings is usually so pronounced that *sorting* can be carried out by hand on a conveyor belt or with an automatic optical system.

The coarsely ground material is often *washed* and subjected to *wet desliming* to remove fine particles. Ore that is intended for direct caustic calcination is separated from fine material (often lamellar silicate particles) by screening.

When the magnesite is not intimately intergrown with its accompanying minerals, lump material for rotary kilns can be sorted by *gravity separation*. This is carried out in one or two stages in a dense-medium separator or cyclone to give particles of sizes of 10–40 or 4–60 mm, respectively. Since the differences in density are small (magnesite 2.95–3.05, dolomite 2.90, calcite 2.70, lamellar silicates 2.5–2.95, quartz 2.65, graphite 2.1–2.2 g/cm^3), separation depends on exact control of the density. Finely divided ferrosilicon is generally used as the dense medium; the impurities are removed as floats. The throughput in modern plants is 50–150 t/h.

If the degree of intergrowth of the magnesite with its accessories lies in the range 0.03–1 mm, separation can only be achieved after thorough grinding of the ore (particle size below ca. 0.2 mm) to expose the accompanying minerals. The most effective separation and enrichment process for this material is *flotation*, which exploits the differing wettabilities of the various minerals

Table 2. Minerals accompanying periclase in sintered magnesia

Mineral	Formula	Abbreviation[a]	C: S[b]	*mp*, °C
Forsterite [*15118-03-3*]	$2\ MgO \cdot SiO_2$	M_2S	0	1890
Monticellite [*14567-83-0*]	$CaO \cdot MgO \cdot SiO_2$	CMS	0.94	1495
Merwinite [*13596-18-4*]	$3\ CaO \cdot MgO \cdot 2\ SiO_2$	C_3MS_2	1.40	1575 (decomp.)
Dicalcium silicate[c] [*10034-77-2*]	$2\ CaO \cdot SiO_2$	C_2S	1.87	2130
Tricalcium silicate [*12168-85-3*]	$3\ CaO \cdot SiO_2$	C_3S	> 1.87	1900 (decomp.)
Magnesioferrite [*12023-16-4*]	$MgO \cdot Fe_2O_3$	MF		1750
Dicalcium ferrite [*12013-62-6*]	$2\ CaO \cdot Fe_2O_3$	C_2F	> 1.87	1435

[a] Abbreviations as used in the cement industry: M = MgO, C = CaO, S = SiO_2, F = Fe_2O_3.
[b] Mass ratio of CaO to SiO_2 (dissolution of small amounts of CaO in the MgO is neglected).
[c] Several modifications.

(→ Flotation). The process must be optimized for each deposit.

The magnesite is finely ground in rod or ball mills and should have as narrow a particle size distribution as possible (ca. 0.09 mm for coarsely crystalline magnesite). This is achieved by combining milling with gravity classification and classification cyclones. Flotation is usually preceded by desliming to remove the finest particles, which would otherwise cause problems in the flotation step.

The removal of talc, $Mg_3(OH)_2Si_4O_{10}$, is straightforward. Upon mixing with air in the presence of weak collectors–frothers (e.g., pine oil and primary aliphatic fatty amines), the talc particles float to the surface on bubbles and are removed by skimming.

The main problem in flotation is the separation of magnesite from dolomite, calcite, and silicates other than talc. Separation is achieved by *direct magnesite flotation* which entails the use of nonspecific, anionic collecting agents (that also act as frothers) and selective depressing agents for dolomite, calcite, and silicates. The flotation cells have a capacity of 0.5–3 m^3. The operational conditions in the flotation cells must be optimized (temperature, pH value, reagent dosing). In large-scale flotation plants, several series of cells are run in parallel. The typical throughput of a series is 20 t/h of magnesite, corresponding to 60 m^3 of suspension with a density of 1.25 g/cm^3. The magnesite concentrate is removed as a foam, then thickened, and brought to a residual water content of 8–12% in centrifuges or filters. The product is then briquetted and burned (see Section 4.3.5). The waste slurries from the desliming and flotation stages are treated in thickeners, filters, and settling basins.

An important sorting process for high-iron sintered magnesia is *magnetic separation* (→ Magnetic Separation). The isomorphic inclusions of iron in the magnesite crystal cannot be removed by mechanical separation. After sintering and subsequent cooling, they are converted to magnetic magnesioferrite ($MgO \cdot Fe_2O_3$), either in the periclase lattice or at its crystal boundaries (see Fig. 3 B). Calcite and dolomite usually do not contain iron in the carbonate lattice; the calcination of large crystals gives large pieces of very pure, usually only lightly sintered, nonmagnetic CaO or CaO and MgO. These oxides can be separated, together with nonmagnetic silicates, from the sintered iron-rich magnesia in magnetic drum separators.

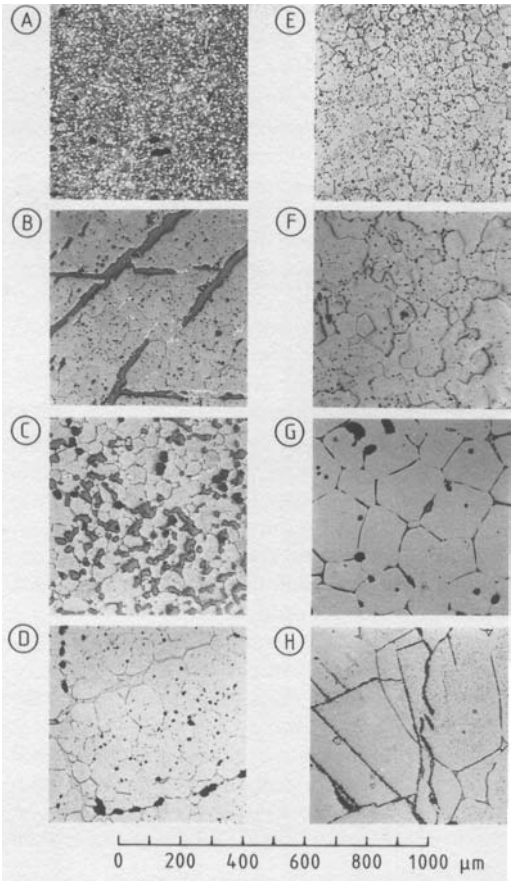


Figure 3. Polished sections of sintered dolomite (A), sintered magnesia (B–G), and fused magnesia (H)
A) Sintered dolomite: white areas are MgO, gray areas CaO, and black areas pores; B) High-iron magnesia sintered in lumps: white domains are 2 $CaO \cdot Fe_2O_3$ and $MgO \cdot Fe_2O_3$, dark gray areas are pores that run parallel to the carbonate cleavage planes (type 4 in Table 3); C) Briquetted high-iron sintered magnesia with a high porosity (type 2 in Table 3); D) Low-iron magnesia sintered in lumps (type 7 in Table 3); E) Seawater magnesia "31" (type 6 in Table 3); F) Sintered magnesia from brine (type 9 in Table 3); G) Low-iron C sintered magnesia (type 8 in Table 3); H) Fused magnesia with periclase cleavage planes

Recently developed magnets are reported to be able to remove pieces of sintered magnesia with a somewhat higher iron content from low-iron sinter [60].

Table 3 lists typical analyses of sintered magnesia produced from natural magnesite (types 1–4, 7).

Table 3. Typical analyses of sintered magnesia

Type of magnesia*	Composition, wt%						Bulk density**, g/cm³	Theoretical density, g/cm³
	MgO	CaO	SiO_2	Fe_2O_3	Al_2O_3	B_2O_3		
1	91.5	1.5	4.5	1.1	1.4	0.01	3.27	3.57
2	89.0	2.4	0.5	7.8	0.3		3.26	3.64
3	88.0	2.5	0.6	8.6	0.3	< 0.01	3.38	3.65
4	91.3	1.9	0.5	6.0	0.3	< 0.01	3.40	3.63
5	98.0	0.7	0.7	0.2	< 0.2	0.30	3.28	3.57
6	97.2	1.9	0.5	≤ 0.2	0.2	0.05	3.43	3.57
7	96.4	2.0	1.1	0.4	< 0.1	< 0.01	3.43	3.57
8	97.2	2.0	0.4	0.3	≤ 0.1	< 0.01	3.42	3.57
9	98.5	0.7	0.15	0.5	< 0.1	0.015	3.45	3.58
10	99.0	0.7	0.1	< 0.1	< 0.1	< 0.01	3.43	3.58

* 1. Lump sintered magnesia with a high silicate (and iron) content
2. High-iron sintered magnesia produced from flotation concentrate
3. Lump high-iron sintered magnesia
4. Lump high-iron sintered magnesia
5. Sintered magnesia produced from seawater type 11 (C : S = 1 : 1)
6. Sintered magnesia produced from seawater type 31 (C : S = 3 : 1)
7. Lump low-iron sintered magnesia
8. Low-iron sintered magnesia with large crystals (LC) produced from flotation concentrate
9. Sintered magnesia produced from $MgCl_2$ brine
10. Sintered magnesia produced from pyrohydrolyzed $MgCl_2$

** According to DIN 51 065, part 2.

4.3.2. Production from Seawater [62, 63]

Few deposits of pure or easily accessible magnesite were known up to the mid-1980 s. In the 1960s the increasing demand for sintered magnesia as a refractory lining material in the steel industry stimulated new developments in methods for production of magnesia from seawater. This method was first introduced on a small scale in 1865 on the French Mediterranean coast. Since the 1930s it has been employed on a large scale in the United States and England (Hartlepool).

The composition of seawater is given in Table 4 [64]. Calculations show that 470 m^3 of seawater are required to produce 1 t of MgO, in practice 600 m^3 are needed. The process is based on the precipitation of magnesium hydroxide (solubility in water 0.0009 wt%) by addition of calcium hydroxide (solubility 0.185 wt%):

$$Mg^{2+} + 2\,Cl^- + Ca(OH)_2 \rightarrow Mg(OH)_2\downarrow + Ca^{2+} + 2\,Cl^-$$

Freshwater from rivers contaminated with suspended material must not flow into the seawater used for magnesia production. Furthermore, the seawater to be extracted should not be mixed with already extracted seawater, this requires a suitable topography and tidal currents that run parallel to the coast. A supply of freshwater (> 40 m^3 per tonne MgO) is required to wash the $Mg(OH)_2$ and to produce the milk of lime. High-purity limestone or dolomite deposits

Table 4. Composition of seawater [64]

Cation	Concentration		Anion	Concentration	
	g/kg	mmol/kg		g/kg	mmol/kg
Sodium	10.47	455.0	chloride	18.97	535.1
Potassium	0.38	9.7	bromide	0.065	0.81
Magnesium	1.28	52.5	sulfate	2.65	27.6
Calcium	0.41	10.2	bicarbonate	0.14	2.35
Strontium	0.013	0.15	borate	0.027	0.44

should be available in the vicinity; they are calcined and slaked to provide $Ca(OH)_2$ as the precipitating agent and should therefore contain minimal quantities of elements that form insoluble carbonates, sulfates, etc. The $MgCO_3$ in dolomite is converted to $Mg(OH)_2$, and thus provides a further source of magnesia.

Production of Precipitating Agent. The limestone or dolomite is thoroughly washed, crushed, and calcined at $> 1400\ ^\circ C$ in a rotary or shaft kiln. The $CaCO_3$ must be completely decarbonated to CaO ($< 0.2\%\ CO_2$), but should be lightly burned, and should contain minimal amounts of sulfur and other contaminants. In order to achieve rapid, complete reaction of the seawater, the calcined lime (quicklime) or dolomite (dolime) is ground, slaked to form $Ca(OH)_2$ and then converted into milk of lime or dolomitic slaked lime (containing ca. 20% solids).

The seawater is pumped to the plant and suspended material is removed. Seawater contains dissolved carbon dioxide and must therefore be treated with sulfuric acid to prevent the subsequent formation of insoluble calcium carbonate on addition of $Ca(OH)_2$:

$$Ca(HCO_3)_2 + H_2SO_4 \xrightarrow{pH4} CaSO_4\downarrow + 2\ H_2O + 2\ CO_2$$

The freshwater is also partially decarbonated.

To ensure maximum removal of carbon dioxide, the water is pumped through a trickling installation into the reactor containing the milk of lime. The water leaving the reactor is pumped back into the sea; it has a low magnesium content, a high calcium content, and a pH of 10.5. In modern plants using the overliming process (see below), the extracted water is first pumped into settling basins where the excess lime is diluted with fresh seawater to precipitate more $Mg(OH)_2$ and to lower the pH to more ecologically acceptable values.

Precipitation. The aqueous suspension of finely precipitated $Mg(OH)_2$ is seeded with larger $Mg(OH)_2$ crystals in large, flat settling tanks to convert it into a thick, dense slurry for filtering. The slurry is removed from the bottom of the tank and fed into tanks where it is washed with decarbonated seawater and decarbonated fresh water to remove most of the alkali metal, chloride, and other ions. The water content of the $Mg(OH)_2$ slurry is then reduced to $< 50\%$ by filtration (vacuum disk or drum filter). The product is used as a starting material for caustic and sintered magnesia (see Section 4.3.5).

Unless specially treated, caustic and sintered magnesia produced from seawater usually contain ca. 0.2% B_2O_3 and small amounts of CaO, SiO_2, Al_2O_3, and Fe_2O_3 derived from the limestone or wastes in the seawater. The contaminants and the $CaO:SiO_2$ (C:S) ratio have a pronounced effect on the behavior of the magnesia during sintering and in its subsequent applications. The C:S ratio is therefore adjusted to the desired value (3–4 or more for the high-quality sinters used in oxygen metallurgy). The B_2O_3 content of the magnesia is also generally lowered to ca. 0.05% by using a 5–12% excess of lime for precipitation (overliming); this increases the pH to 12 and minimizes the adsorption of boron. Addition of small quantities of Na_2CO_3 prior to sintering further decreases the B_2O_3 content to 0.03% due to the formation of volatile sodium borate during sintering. Prolonged sintering at the maximum temperature decreases the B_2O_3 content. Ion exchange can also be used to lower the boron content of the seawater and thus of the sintered product.

Typical compositions of the two most common types of magnesia sinter derived from seawater are given in Table 3 (types 5 and 6).

4.3.3. Production from Brines

Precipitation of Magnesium Hydroxide. Large marine and terrestrial salt deposits were formed in various geological periods. Fossil mineral salt deposits have been mainly mined for their potassium salts: magnesium salts or magnesium brines are obtained as waste or as a byproduct.

Magnesium salts are formed in arid areas by evaporation of brines in flat basins, their composition often differs from those obtained from seawater. Magnesia is obtained from magnesium (potassium) brines of varying origin:

1. Synthetic brines produced during solution mining (e.g., Veendam, Netherlands)
2. Spent liquors obtained in salt production from natural brines (e.g., Dead Sea, potash production; Laguma de Rey, Mexico, sodium sulfate production)

3. Spent liquors obtained in salt production from salt deposits (e.g., Teutschenthal, Germany)

In the Netherlands (Veendam, Ost-Groningen), purified freshwater is forced into an anticline at a depth of 1500 m to dissolve the bischofite ($MgCl_2 \cdot 6\ H_2O$) and carnallite ($KCl \cdot MgCl_2 \cdot 6\ H_2O$) [65]. The brine is then pumped to the surface and contains 20 wt% $MgCl_2$ and small amounts of alkali chlorides and $MgSO_4$. After removal of the sulfate, the magnesium is precipitated as $Mg(OH)_2$ with dolomitic slaked lime (see Section 4.3.1), thickened, and then sintered. The advantage of this deposit is that the bicarbonate and boron contents of the resulting brines are much lower than those of seawater and do not have to be removed by complex methods.

Pyrohydrolysis of Magnesium Chloride [66, 67]. In pyrohydrolysis, metal (M) halides (especially chlorides) are thermally decomposed in the presence of superheated steam at 300–1000 °C to give a pure, fine metal oxide precipitate:

$$M(II)Cl_2 + nH_2O \rightarrow MO + 2\,HCl + (n-1)H_2O$$

Pyrohydrolysis was originally used to recover hydrochloric acid in the pickling of steel but it can also be employed for the large-scale production of metal oxides. Magnesium oxide is produced from natural or synthetic $MgCl_2$ brines. Complete hydrolysis of $MgCl_2$ by conventional methods is not possible because evaporation leads to precipitation of bischofite ($MgCl_2 \cdot 6\ H_2O$) which, on further heating, melts or dissolves in its own water of crystallization. Complete pyrohydrolysis of $MgCl_2$ is achieved in a spray reactor without hydrolysis of $CaCl_2$ and alkali chlorides (Aman process).

The *Aman process* has been used since 1973 in an industrial plant with a capacity of 70 000 t/a near the Dead Sea. The starting material for this process is the spent liquor (brine) obtained after extraction of KCl from a magnesium- and alkali-chloride-rich natural brine. The compositions of these brines are given in Table 5.

The brine is purified to remove bromide and traces of boron and then fed via steel pipes into the spray nozzles of the reactor. It is sprayed into the cylindrical, externally insulated reactor at ca. 600 °C. The reactor is lined with refractories, it also has a conical base and tangentially positioned burners which ensure rotating circulation of the reactants. The exhaust gases leave the top of the reactor at 450 °C. The water evaporates from the atomized brine droplets leaving a perforated chloride crust which reacts with the steam to form MgO and HCl. The hollow spherical MgO agglomerates are removed from the bottom of the reactor via a gas lock, together with non-pyrolyzed, water-soluble potassium, sodium, and calcium salts.

Table 5. Composition (in grams per liter) of brines used in the production of magnesia by the spray roasting method

Component	Natural brine	Brine for pyrohydrolysis after KCl extraction
Mg^{2+}	41	102
Na^+	40	2
Ca^{2+}	17	9
K^+	7.5	2
Cl^-	215	319
Br^-	5	6
SO_4^{2-}	0.7	
Total	326	440
$MgCl_2$	163	400

The crude product is washed with water and hydrated in a stirred tank, and then concentrated in a thickener. The resulting slurry is difficult to filter and is washed and dewatered in a two-stage vacuum drum filter. The calcined product typically contains $\geq$ 99.5 wt% MgO, $<$ 1 wt% CaO, $\leq$ 0.05 wt% SiO_2, $\leq$ 0.05 wt% Fe_2O_3, $\leq$ 0.005 wt% Al_2O_3, and $\leq$ 0.01% B_2O_3; its specific surface area is 2–50 m^2/g, the loose bulk density ranges from 0.8 to 0.2 g/cm^3.

Most of the MgO dust in the hydrogen chloride exhaust gas is removed with a cyclone. The hot gas is then used to heat the fresh $MgCl_2$ solution in a wash recuperator where any remaining MgO dust is collected. The HCl gas is scrubbed with water in an absorption column. After removal of the combustion gases and steam, 20% hydrochloric acid is obtained which is used to dissolve local phosphate deposits; this makes the process economically viable.

Pyrohydrolysis of Magnesium-Containing Waste. The principle of the above method can also be used for the pyrohydrolysis of chloride solutions obtained from impure magnesium minerals such as magnesite, tailings, dust, asbestos,

talc, and other magnesium silicate waste [68]. The hydrochloric acid byproduct is recycled within the process (former Yugoslavia, Nepal, Austria, and Germany).

The magnesium is dissolved with hydrochloric acid in a two-stage process. Neutralizing agents (filter dust or crude MgO), oxidizing agents (air, chlorine gas), and sulfate ions are then added. Heavy-metal hydroxides, calcium sulfate, and other contaminants precipitate out and are filtered off to leave a $MgCl_2$ solution that is concentrated in a vacuum evaporator and sprayed into the reactor.

4.3.4. Miscellaneous Processes

Sulmag II Process [69]. The Sulmag II process was developed by Sulzer for producing light-burned caustic magnesia in a gas suspension kiln from low-magnesite ores. Dissolved magnesium chloride is obtained by selective extraction with recycled NH_4Cl solution:

$$2\,NH_4Cl + MgO + H_2O + \text{Contaminants} \rightarrow 2\,NH_4OH + MgCl_2 + \text{Tailings}$$

After filtration and washing, needle-shaped crystals of nesquehonite are precipitated out in the reactor and filtered off:

$$MgCl_2 + (NH_4)_2CO_3 + 3\,H_2O \rightarrow MgCO_3 \cdot 3\,H_2O\downarrow + 2\,NH_4Cl$$

Caustic magnesia with a high specific surface area is obtained by directly heating the nesquehonite with oil or gas and briquetting while still hot. The product is then sintered. This process is reported to produce significant energy savings and is environmentally favorable due to the almost complete recycling. A pilot plant has been in operation in Switzerland for several years and a plant designed to produce 30 000 t/a of sintered magnesia is under construction for Tamilnadu Magnesite in India.

Other Processes. Processes have been described in which dolomite is decarbonated to form a mixture of CaO and MgO and the CaO is then chemically extracted. Patents also describe the partial decarbonation of dolomite, slaking of the MgO, and recarbonation. The product consists of large nesquehonite crystals ($MgCO_3 \cdot 3\,H_2O$) which are separated from the microcrystalline $CaCO_3$ by flotation. The accessory calcite and dolomite can also be extracted from magnesite rocks because they are readily soluble in acid.

Magnesia can also be prepared from Epsom salt ($MgSO_4 \cdot 7\,H_2O$) or kieserite ($MgSO_4 \cdot H_2O$) but this technique is not used industrially at present.

MgO and CaO Production by Sintering of Dolomite. Complete decarbonation of dolomite theoretically leaves a finely crystalline mixture of MgO (42%) and CaO (58%).

Since large supplies of natural, high-purity dolomitic rocks are available, removal of contaminants by beneficiation is not economical. The high-purity lump dolomite is calcined, and sintered in a rotary or shaft kiln. In a special procedure the dolomite is decarbonated, briquetted, and sintered at a lower temperature than the lump material to give a high-density product.

The CaO and MgO crystals in sintered dolomite lie directly adjacent to one another (Fig. 3 A); despite the intense burning they are much smaller than those of sintered magnesia because they mutually hinder one another's crystal growth.

4.3.5. Burning of Caustic and Sintered Magnesia

Caustic Magnesia is a very reactive, finely crystalline material that is produced by burning $MgCO_3$ or $Mg(OH)_2$ slightly above the decomposition temperature. The decomposition curves for $Mg(OH)_2$, $MgCO_3$, and $MgCO_3 \cdot 3\,H_2O$ are shown in Figure 4. Caustic magnesia may be the end product of magnesia production (for uses, see Section 4.4.3) or may be further burned to give sintered magnesia.

Caustic magnesia is produced industrially by calcining lump $MgCO_3$ (up to 50 mm) or finer material at 600–1000 °C in shaft, rotary, or multiple-hearth (Herreshoff) kilns. Heat-exchange kilns are also employed. In the case of $Mg(OH)_2$, dewatered filter cakes are usually calcined in lump form at ca. 950 °C in multiple-hearth kilns. These kilns generally contain ten shelves

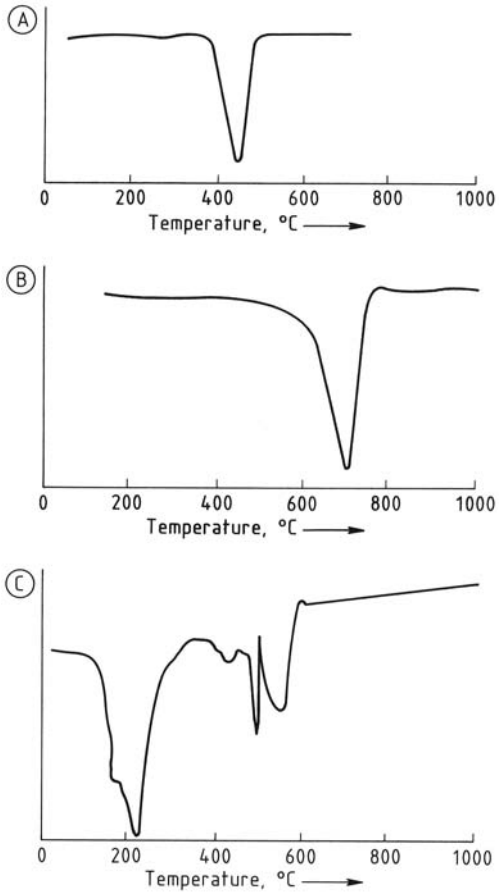


Figure 4. Differential thermal curves for $Mg(OH)_2$ (A), $MgCO_3$ (B), and $MgCO_3 \cdot 3\ H_2O$ (C) [81]

(hearths), one above the other; each hearth is provided with four burners in the vertical cylindrical kiln wall. The material is fed continuously into the top of the kiln and its residence time can be adjusted via the rake that rotates above each hearth. The calcining conditions must be carefully adapted to the contaminants in the feed, otherwise overburning results in excessive growth of the reactive MgO crystallites which lowers their activity. In modern gas suspension furnaces, $Mg(OH)_2$ or flotation concentrate can be converted into homogeneous caustic magnesia; energy consumption is low [70].

Caustic magnesia was formerly produced exclusively from cryptocrystalline magnesite with a low iron content but is now also obtained from all types of magnesite and $Mg(OH)_2$. Its MgO content ranges from ca. 65 to 99 wt%, and may even reach 99.9%. The magnesia is often ground prior to use. Extremely reactive caustic magnesia may have a surface area of up to 160 m^2/g.

Depending on the burning temperature, the product is termed *light burned* (870–1000 °C) or *hard burned* (1550–1650 °C). Light-burned caustic magnesia becomes hydrated in cold water and is soluble in dilute acid. It has a loose bulk density of 0.3–0.5 g/cm^3 and a specific surface area (BET) of 10–65 m^2/g. Hard-burned caustic magnesia has a loose bulk density of 1.2 g/cm^3 (bulk density 2 g/cm^3).

Sintered Magnesia. Most (ca. 85%) $MgCO_3$ and $Mg(OH)_2$ is converted into sintered magnesia (also known as magnesia clinker or dead-burned magnesia, Fig. 3 B–G). Typical analyses and densities of sintered magnesia are given in Table 3). Uses are described in Sections 4.4.1 and 4.4.2. Sinter quality depends not only on chemical composition but also on the bulk density or porosity; a high density gives a better product. Sintering capability decreases considerably with increasing purity, making a high density difficult to achieve. In the early 1960s lump magnesite with a low iron content could not even be sintered to a density of 3.40 g/cm^3 (porosity ca. 5 vol%). Although some briquetted, low-iron, C_2S-containing sinters have densities of up to 3.46 g/cm^3 (porosity ca. 3.5 vol%), the density is usually in the range 3.35–3.40 g/cm^3. The final bulk density of the sinter depends on chemical composition, sintering temperature and time, and the degree of compression of the material prior to sintering (briquetting is therefore used and not pelletizing or granulation). The material is usually sintered in lump form.

Caustic magnesia (preferably hot, 200–500 °C) is compressed into briquettes on high-pressure roller presses (pressure 4–15 t/cm). The briquettes are then screened and dried or hardened. The screened fines are mixed with fresh caustic magnesia and briquetted.

Binding and adhesive agents are added to fine *magnesite flotation concentrates* prior to briquetting; magnesium sulfate solution, sulfuric acid, lignin sulfonates, and reactive MgO flue dust are usually used. Magnesite concentrates are also used to produce double-burned sinter. Here the concentrates are first calcined to reduce the mass and the resulting gas pressure; they are then briquetted and sintered.

In some cases dewatering of *magnesium hydroxide slurries* is so efficient (high-performance filter presses) that pieces of the hard filter cake can be fed directly into a rotary kiln. This saves a considerable amount of energy and the sintered magnesia product usually has a sufficiently high bulk density. In most modern plants, however, the hydroxide slurry is first calcined and then briquetted.

Shaft kilns are heated with coke, gas, or oil (with or without oxygen) and must be fed with large pieces of material due to the pattern of gas circulation and movement of material within the shaft. Today, shaft furnaces can be better controlled than was the case a few years ago [71]. Smaller lumps (0.5–60 mm, product 0–30 mm) are used in rotary kilns to minimize abrasion and guarantee even movement of the material from the rotating kiln wall. Rotary kilns are generally oil- or gas-fired, sometimes with addition of coal dust or petroleum coke. Depending on the content of fluxing agents, sintering is performed at 1700 to $>$ 2000 °C for 2–15 h (rotary kiln) or ca. 20 h (shaft kiln). The sinter from rotary kilns is generally homogeneous but has a smaller particle size and contains more dust than that from shaft kilns; this can lead to undesirable, round particles and an excess of fines during brick production. Cryptocrystalline magnesite with a low content of fluxing agents gives a sinter with a needle-like structure also after sintering in a rotary kiln.

Energy consumption for the burning of high-quality sintered magnesia from various starting materials depends on kiln design and the water content of the $Mg(OH)_2$ slurry; typical values are as follows (MJ/kg MgO):

Magnesite: single-stage burning (magnesite → sinter)	4.5–12
Magnesite: double-stage burning (magnesite → caustic magnesia → sinter)	8–16
Seawater magnesia [$Mg(OH)_2$ → caustic magnesia → sinter, value includes energy required for production of precipitation agent]	22–32
Sulmag II process	23
Magnesia produced by pyrohydrolysis	40

In an effort to improve sinter quality, manufacturers first turned their attention to increasing the size of the periclase crystallites by adding sinter-promoting agents, such as ZrO_2. Burning conditions were subsequently modified. In larger crystallites the MgO is less easily reduced by carbon at high temperatures (e.g., in magnesia–graphite bricks; see also Fig. 3 G).

Modification of Sinter Composition. In the case of concentrates, slurries, or caustic magnesia, the final composition of the sinter can be modified by mixing in small amounts of additives such as CaO, SiO_2, and Fe_2O_3 prior to briquetting. Magnesite concentrates obtained by flotation and fine $Mg(OH)_2$ or MgO can also be mixed with large quantities of other fine substances (primarily chromium ores, calcined lime or dolomite), briquetted, and sintered to form coclinker. In refractory bricks made from pure sintered magnesia and low-silicate chromium ores, the chromium ore only becomes fully integrated with the periclase at high temperature ($\geq$ 1700 °C) and in a suitable atmosphere. In briquettes made from magnesia and fine chromium ore, however, sintering results in complete dissolution of the mixed spinels of the ore in the periclase. On cooling, fine crystalline spinels or skeleton crystals precipitate in the periclase; some idiomorphic spinels $(Mg, Fe^{2+})O(Cr, Fe^{3+}, Al)_2O_3$ are evenly deposited between the periclase crystallites.

4.4. Uses

The use of magnesium oxide and hydroxide as intermediates in the production of magnesium metal is described elsewhere (→ Magnesium).

4.4.1. Use of Sintered Magnesia as a Refractory

Oxides are best suited as "vessel linings" for most industrial high-temperature processes. Magnesium oxide has the highest melting point of the moderately priced oxides and is therefore an important raw material for refractory bricks and other materials (→ Refractory Ceramics). It is the only material apart from ZrO_2 that can withstand long-term heating above 2000 °C. Refractory construction materials based on magnesia (with chromium ore or CaO) are also resistant to neutral and basic slags and gases.

The boom in the production of low-iron sintered magnesia occurred after it was discovered that the

use of refractory magnesia linings in steelmaking vessels (open-hearth furnace, electric arc furnace, and basic oxygen furnaces) allowed phosphorus and sulfur to be removed in the basic slag.

Sintered magnesia is converted into fired bricks by techniques used for traditional ceramics. It is first broken up, ground, and classified. Batches (1 to > 2 t) of individual fractions are mixed with a temporary aqueous binder (e.g., lignin sulfonates) and pressed into bricks of the desired shape weighing up to ca. 35 kg. After drying, the bricks are usually fired at 1500 –1800 °C in a tunnel kiln. They have a final porosity of 12–20 vol%.

Bricks (especially those that are to be used in basic oxygen furnaces) may then be *impregnated with pitch* to prevent the slag from infiltrating them and dissolving their surface. Fired magnesia bricks have a high thermal expansion coefficient and a relatively low thermal shock resistance, especially during periodic operation. Fired bricks made from sintered magnesia and chromium ore (or alumina or spinel) are more resistant to thermal shock.

A large proportion of the magnesia bricks are not fired (i.e., do not form a ceramic bond) but are *bonded with pitch or resin*. The individual particle fractions are mixed with hot pitch, compressed into bricks, and tempered at 200 – 500 °C. When the bricks are used, a coke structure is formed which makes them more elastic than the ceramic-bonded bricks. Addition of 5–20% flake graphite increases heat transfer, lowers thermal expansion and attack by the slag, and improves thermal shock resistance. This development is applied in electric arc furnaces, converters, and metal ladles for secondary steelmaking processes. The low-oxygen atmosphere formed in these bricks during use can cause significant reduction of the magnesia. To prevent this, manufacturers have developed high-quality magnesia with larger periclase crystals and a low internal sinter surface area [72].

Chemically Bonded Bricks with chrome ore contain chemical bonding agents (e.g., Epsom salt, $MgSO_4 \cdot 7\ H_2O$) and are produced in the same way as fired magnesia bricks up until the drying stage. The bonding agents decompose and form a ceramic when the bricks are used at high temperature.

Pig iron mixers, transport vessels, and secondary refining ladles used in the steel industry are lined with magnesia bricks. Sliding gates and nozzles are also made from sintered magnesia.

On account of their high refractoriness (refractoriness under load and low creep in compression), resistance to alkali, and high heat storage capacity, magnesia bricks are used in the crowns, port linings, and the hot parts of the checkerwork in glass-melting tanks. Magnesia–chrome or magnesia–spinel bricks are more suitable for the sintering zones of rotary cement kilns due to their good stress deformation behavior. Magnesia bricks are preferred for lime shaft kilns. Large quantities of sintered magnesia are used for unshaped repair materials.

The increased demands made on refractory materials as a result of higher operating temperatures and shorter tap to tap times in metallurgical furnaces and reactors can only be met by pure, high-density magnesia sinters. Small quantities of "contaminants" are disadvantageous if they form low-melting eutectics with MgO (e.g., with CMS at 1485 °C or with C_2F at 1200 °C, Table 2) because this leads to deterioration of mechanical properties (e.g., strength and volume stability) at high temperatures. High-quality sinters therefore have a low content of high-melting silicate phases (such as C_2S), a low B_2O_3 content, and a high degree of direct periclase–periclase contact (without intermediate silicate phases).

4.4.2. Use of Sintered Magnesia as a Heat-Storage Medium

Magnesia bricks have a high heat storage capacity (about 160% that of chamotte) and a high thermal conductivity [73]. They are used in efficient off-peak storage heaters. The heat generated by a heating element is transferred to the magnesia brick and increases its temperature.

The most suitable bricks are fired and have the following properties: a high bulk density (up to 3.10 g/cm^3) and thus a high degree of sintering, a high thermal conductivity (up to 9 W m^{-1} K^{-1} at 600 °C), and a high specific heat (1.1 kJ kg^{-1} K^{-1} at 600 °C). The influence of impurities on these factors is not as critical as in refractories.

The bulk density depends on the chemical composition, the sinter porosity and the method used to produce the bricks. Thermal conductivity is increased by a high periclase content and a low porosity. The specific heat is only slightly

lowered by SiO_2 and Al_2O_3, but is significantly lowered by CaO, Cr_2O_3, and Fe_2O_3. The bricks should not contain free CaO (risk of hydration) or crystal phases with different modifications.

4.4.3. Uses of Caustic Magnesia

Agriculture. In the 1970 s and 1980 s the use of caustic magnesia increased the most in fertilizers and animal feeds [74–77]. Magnesium is a constituent of plant chlorophyll and also plays a role in enzyme activity. It is regarded as an essential plant nutrient and is used in *fertilizers* (→ Fertilizers, 1. General). Magnesium deficiency (symptoms: chlorosis, necrosis) became especially common after fertilizers containing a high percentage of potassium replaced potassium –magnesium fertilizers based on kainite ($KCl \cdot MgSO_4 \cdot 3\ H_2O$). In countries with salt deposits (e.g., Germany), magnesium is supplied in the form of kieserite, magnesia lime (> 10% MgO), and calcined dolomite. In all other countries (e.g., Netherlands, UK, USA), caustic magnesia (particle size 4 mm, ≥ 70% MgO) and magnesite are used.

Ground caustic magnesia (particle size ≤ 2 mm, ≥ 85% MgO) is used as a *supplement for animal feeds*. Magnesium is usually present in green fodder, but cattle and sheep feeding on fodder from slightly acid soils may suffer from "grass staggers" in spring which is caused by magnesium deficiency.

Building Industry. Caustic magnesia mixed with concentrated solutions of magnesium salts and a small amount of sodium phosphate becomes extremely hard and sets in air (e.g., with $MgCl_2$ solutions it forms $3\ MgO \cdot MgCl_2 \cdot 11\ H_2O$). However, the hardened material is not stable in water and is therefore not a true cement (which must set in air and water) but a *magnesia binder*. These binders were originally referred to as Sorel cement or as stonewood (xylolith) when mixed with sawdust as a filler. Since their high chloride content can corrode metals and other construction materials, magnesia binders must never be reinforced. They can be used as mastics or protective coatings at room temperature. Large quantities of caustic magnesia derived from carbonate are used for stonewood or plaster flooring in the Eastern Bloc. The low-iron caustic magnesia generally used for this purpose contains 75–87% MgO and is ground (40 – 45% < 10 μm).

An important use of caustic magnesia is in *lightweight construction boards* (Heraklith) for thermal and acoustic insulation (thickness 5 – 100 μm). The boards are produced on fully automatic lines at ca. 60 °C from finely ground caustic magnesia (MgO content 65–80%, starting material magnesite containing silicates and dolomite), $MgSO_4$ solution as a binder, and impregnated pinewood chips. Mineral wool is also used as a filler. Subsequent hardening results in highly nonflammable boards which can be made fire-retardant and resistant to moisture and microorganisms by coating one side with plaster. The prefabricated boards are used to make formworks for concrete, and large quantities are used in combination with polystyrene in laminated boards.

Other Uses. Caustic magnesia containing ca. 70–99% MgO is used in wastewater treatment, in prevention of corrosion, as a buffer, in preparative chemistry, and in the pharmaceutical industry.

Caustic magnesia can be used to remove heavy metals and silicate from wastewater. Ammonia can be precipitated and removed in the form of $Mg \cdot NH_4 \cdot PO_4 \cdot 6\ H_2O$ following addition of phosphoric acid and caustic magnesia [78]. The magnesium ammonium phosphate can be used as a fertilizer. This principle is also used to remove phosphate. Caustic magnesia is a mild base and neutralizes acids and acidic water, it also acts as a weak buffer.

Low-iron caustic magnesia is used as a filler in the plastics and rubber industries and allows adjustment of viscosity and stiffness. It also acts as a chlorine accepter in chlorinated hydrocarbon polymers and stabilizes them.

Large quantities of pure magnesium oxide are used in the chemical industry for the synthesis of magnesium compounds. Basic magnesium sulfonates and finely dispersed MgO are used as lubricant and motor fuel additives to neutralize combustion residues. Magnesium oxide in fuels binds sulfur and other compounds and prevents formation of acidic exhaust gases.

Large quantities of caustic magnesia are used in the paper and cellulose industries (magnesium bisulfite digestion). Caustic magnesia with graded particle size is also used as a polishing agent.

Caustic magnesia, magnesium carbonate (precipitated from caustic magnesia), and high-purity magnesia are utilized in preparative chemistry and cosmetics. In the pharmaceutical industry magnesium oxide ($\geq$ 98% MgO) is used as an antacid, as a powder base, and for magnesium substitution therapy.

A potential use of caustic magnesia is as a slag additive to reduce the dissolution of MgO from refractory bricks in the steel industry. Treatment of the surface of grain-oriented steel transformer sheets decreases magnetic losses. A suspension of caustic magnesia is applied to the sheet and annealed.

4.4.4. Uses of Magnesium Hydroxide

The most important use of magnesium hydroxide is as an intermediate in the production of magnesia. In Japan it is used to replace polyhalogenated diphenyl ether as a flame retardant in thermoplasts processed at 200–350 °C. Hydrated alumina cannot be used at these temperatures because of its low decomposition temperature. This application is being tested in Germany [79]. Well-dispersed single $Mg(OH)_2$ crystals seem to be more suitable than agglomerates.

4.5. Economic Aspects

The magnesia market is characterized by continuous ups and down. The development in the early 2000s was as follows:

Until mid-2008 there was a world magnesia shortage because of continued global growth in the main consuming industries. China dominated the global market with its exports covering about one-forth of the market outside China [80]. Exports fell substantially in 2009 as a consequence of reduced demand after the economic decline and because of the export taxes imposed on MgO products by the Chinese government.

4.6. Fused Magnesium Oxide

4.6.1. Properties

Fused magnesia is crystalline magnesium oxide (*mp* 2800 °C) that has been melted in an electric arc. It has the same crystal structure as sodium chloride (i.e., face-centered cubic) and occurs naturally as periclase. No phase changes take place on heating up to the melting point.

Pure fused magnesia is white, although the presence of iron impurities impart a greenish color due to the presence of the mixed crystal compound magnesiowüstite (Mg,Fe)O. When the pale green fused magnesia is heated in air, the color changes to brown owing to the formation of magnesioferrite ($MgFe_2O_4$) [82].

Fused magnesium oxide has a much lower tendency to undergo hydration than sintered magnesium oxide or caustic calcined magnesite [83], and is essentially stable towards the atmosphere. Thus, polished crystal faces of periclase only begin to lose their luster after several weeks, depending on atmospheric moisture content and temperature. Reaction with atmospheric carbon dioxide to form basic carbonates is also noticeable only after several weeks or months.

Fused magnesia is stable in an oxidizing atmosphere up to ca. 2200 °C, above which vaporization and partial dissociation take place. Theoretical vapor pressures of magnesium oxide and partial pressures of its dissociation products in the temperature range 1227–2727 °C are given in [84].

In a reducing atmosphere (e.g., in the presence of carbon), fused magnesia is stable only up to ca. 1700 °C. This temperature is also the limit for vacuum use because of the resultant vapor pressure [85].

The most important physical properties of fused magnesia (e.g., thermal conductivity and electrical resistivity) depend on the density, purity, particle size distribution, type, and distribution of the foreign mineralogical phases present. Consequently, data obtained from measurements on single magnesium oxide crystals [86] or pure, polycrystalline magnesium oxide (> 99%) cannot be regarded as characteristic of commercial fused magnesia.

The following data (see also Chap. 4) are therefore partly based on information from manufacturers of fused magnesia and include the grades most commonly available (MgO content > 92%, particle size 0–400 μm):

Density of pure periclase (theoretical)	3.56–3.58 g/cm^3
Knoop hardness	370
Mohs hardness	6
Linear thermal expansion coefficient (1000 °C)	14×10^{-6} K^{-1}

(*Continued*)

Specific heat (c_p)	
100 °C	0.96 kJ kg^{-1} K^{-1}
1800 °C	1.21 kJ kg^{-1} K^{-1}
Thermal conductivity (1000 °C) for densities of	
3.0–3.58 g/cm^3	ca. 2.9–8.4 W m^{-1} K^{-1}
Dielectric constant, (30 Hz–1 MHz)	11
Electrical resistivity (technical grades, manufacturers' data)	
600 °C	$2\times10^9 - 10^{11}$ Ω · cm
800 °C	$3\times10^8 - 2\times10^9$ Ω · cm
1000 °C	$10^7 - 9\times10^7$ Ω · cm
Electrical resistivity of MgO single crystals (for comparison)	
500 °C	2×10^{13} Ω · cm
1000 °C	6×10^8 Ω · cm

4.6.2. Production

Fused magnesia is mainly produced from naturally occurring magnesite ($MgCO_3$) that has been calcined at 1200–1400 °C ("dead burned"). The calcined product usually contains > 92% MgO, 1–4% SiO_2, 0.5–2% CaO, 0.1–2% Al_2O_3, up to 0.2% Fe_2O_3, and traces of ZrO_2, NiO, Na_2O, K_2O, B_2O_3, and S. Magnesite from seawater (see Section 4.3.2), containing > 97% MgO, is also used as a raw material.

When fused magnesia is used as an electrical insulator for heating applications (tubular elements), the amount of conductive impurities in the raw material must be minimal. Therefore magnesites that contain, for example, iron oxides or sulfur in appreciable amounts cannot be used. Contamination with sulfur is avoided by calcining the magnesite in wood-fired furnaces.

Fused magnesia is usually produced in a batch process by melting the raw material at 2800–3000 °C in electric arc furnaces [87].

The furnaces are heated by either single-phase or, more usually, three-phase a.c. current (Higgins furnace, block-making process). They have a moving base and a conical, water-cooled steel cover. The raw material is added batchwise and acts as a refractory lining for the furnace walls, and the furnace base is usually lined with magnesite bricks or a rammed graphitic material bonded with bitumen. The furnaces are heated by either two or three graphite electrodes. The arcs are struck onto a layer of coke and the electrodes can be vertically adjusted to suit the height of the molten bath.

The current in, for example, a 1000-kW a.c. electric arc furnace (Higgins furnace) is 6000–9000 A at 90–150 V. The electric power required to produce the fused blocks is ca. 2000–4000 kW · h/t [87].

In addition to the traditional block-making process (Higgins furnace), tilting furnace technology is also used for the industrial production of fused magnesia [88]. The melting process is continuous, and the molten product is poured intermittently into water-cooled or refractory-lined molds. Manufacture of fused magnesia by an electric arc trough melting process [89] and in a high-frequency plasma [90] has also been described.

On completion of melting, the electrodes are withdrawn and, after a cooling period, the furnace cover is removed. The fused magnesia blocks can weigh as much as 20 t. After cooling to room temperature, preliminary size reduction is carried out. This is followed by further size reduction of the large broken pieces. The blocks are not homogeneous with respect to structure or chemical composition, and their surface is covered with a coating or crust of agglomerated sintered magnesia.

The impurities in the raw product consist of compounds formed from the quaternary system $MgO-CaO-SiO_2-Fe_2O_3$, e.g., magnesium silicates such as forsterite ($2\,MgO \cdot SiO_2$, *mp* 1890 °C) and monticellite ($CaO \cdot MgO \cdot SiO_2$, *mp* 1500 °C, incongruent melting) whose melting points lie far below that of pure magnesium oxide. The foreign phases are mainly concentrated locally in the surface zone of the fused block and constitute the "melt residues" along with imperfectly melted magnesium oxide (the sintered crust). They are hand separated from the fused product which is a white to pale green crystalline periclase with crystallite sizes from several hundred micrometers up to several centimeters. Hand sorting is the normal industrial method of impurity removal, no better method has yet been found. However, some of the silicate impurities associated with the raw material form inclusions of foreign phases in the crystals of the pulverized magnesium oxide end product. This is because the rate of growth of the magnesium oxide crystals in certain regions of the solidifying melt is greater than the migration rate of the melt residues. Thus, magnesium oxide blocks from a Higgins furnace contain foreign phases (monticellite and forsterite) deposited at the grain boundaries, while magnesium oxide produced by casting from a tilting furnace usually exhibits

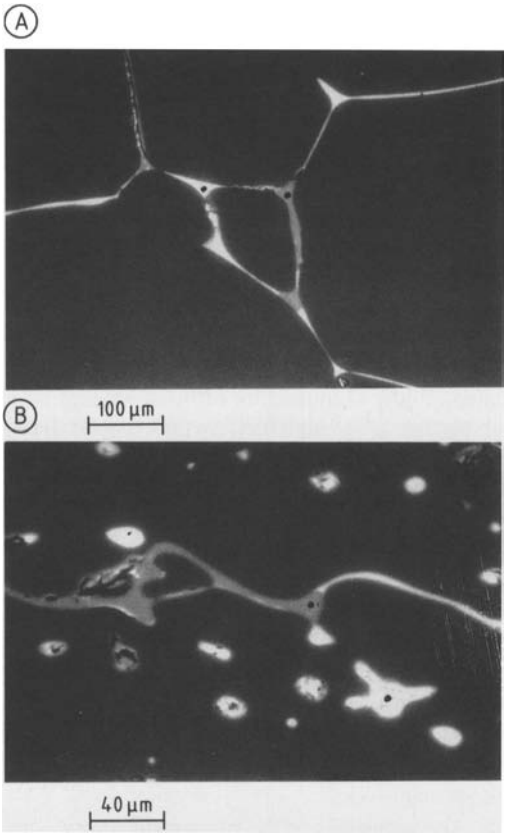


Figure 5. Photomicrograph of polished sections of fused magnesia (dark areas: MgO; light areas: foreign phases monticellite and forsterite)
A) Magnesia produced in a Higgins furnace: foreign phases deposited at the grain boundaries; B) Magnesia produced by casting in a tilting furnace: foreign phases occur mainly as intracrystalline material

an intracrystalline distribution of these phases (see Fig. 5). Further treatment of the hand-sorted, size-reduced, fused product includes several size reductions with jaw crushers, roller crushers, and gyratory crushers. Iron particles produced by abrasion and other magnetic impurities are removed by high power magnetic separators. Finally, the finely grained material is classified into particle size ranges using a series of sieves. Fused magnesia intended for electrical heating applications is heated in an oxidizing, neutral, or reducing (although not carbon-forming) atmosphere at 1100–1400 °C [91] to remove moisture and carbon (originating from breakdown of the electrodes) and to reduce surface and lattice defects of the magnesium oxide particles.

4.6.3. Uses

Fused magnesia is mainly used as an *insulating material* in the electrical heating industry (→ Insulation, Electric). The particulate material is packed into the space between the heating coil and the outer tube in a heating element for air or liquids [92]. Such elements are used in IR heaters, grill elements, tubular hot plates, and tubular heating elements for ovens, storage heaters, radiators, continuous-flow heaters, washing machines, coffee machines, electric irons, and industrial liquid heating equipment.

The most important factor in this application of fused magnesia is its combination of high electrical resistance with high thermal conductivity. This is superior to that of other refractory oxides or other materials.

The grade of magnesium oxide recommended depends on the application for which the heating element is used (surface temperature of the tube) and the electrical insulation requirements. The grades usually have a size range of 0–400 µm and a tap density of 2.2–2.7 g/cm^3. The product must have well-defined flow properties, which can be influenced by modifying the particle shape by treatment in a fluidized bed [93].

The electrical and thermal properties of fused magnesia can be improved by adding synthetic or natural minerals (e.g., pyrophyllites [94], fused zircon sand [95], enstatite [96], cordierite [97], or magnesium aluminum silicates [98]) to the prepared mixture of magnesia particles. The mechanism of action of these additives has not yet been explained.

The chemical composition of some of the industrial grades of fused magnesia (electrical grade) are given in Table 6. The important properties of the mixture of fused magnesia particles (as used for electrical heating, i.e., thermal conductivity, long life time, sinter index, specific electrical impedance, leakage current, tap density, flow rate, grain size distribution) are tested by standard methods (ASTM D 2858, D 2900, D 3026, D 3215, D 3347; DIN 44 872).

Fused magnesia has a further application as a *refractory material*, especially as a lining for induction furnaces and crucibles. Induction furnace linings based on fused magnesia do not shrink (unlike those based on sintered magnesia) owing to the low porosity of the fused material. Magnesia–graphite bricks with an addition of

Table 6. Chemical composition (typical) of some commercially available fused magnesia grades

Producer	Grade	MgO, wt%	SiO_2, wt%	CaO, wt%	Fe_2O_3, wt%	CaO : SiO_2 ratio	Density, g/cm^3
Pechiney, France	CA 250[a]	95.5 (min.)	2.2	1.5	0.008	0.68	2.39[c]
Hüls, FRG	Dynatherm[a] 1246 CS	96.3	2.0	1.5	0.10 (max.)	0.75	2.36–2.40[c]
Thermal Syndicate, UK	Maglox[a] 1GN1	96.8	2.3	0.8	0.10	0.35	2.35–2.40[c]
Universal Abrasives, UK	M 70[a]	93.1	3.7	1.6	0.16	0.43	2.36–2.40[c]
Muscle Shoals Minerals, USA	22 SR[a]	96.0	3.1	0.8	0.07	0.26	2.38–2.41[c]
Pechiney, France	UR1 AT[b]	98.0	0.5	1.1	0.15	2.2	3.50[d]
Baymag, Canada	Electromag[b]	96.7	0.2	2.2	0.5	11	3.50[d]
Tateho, Japan	KMA[b]	98.5	0.35	0.9	0.15	2.6	3.57[d]

[a] Electrical grades for high-duty tubular heating elements.
[b] Refractory grades.
[c] Tap density according to ASTM D 3347, grain size 40–325 mesh.
[d] Apparent density.

fused magnesia [99] are used in ultrahigh-power furnaces for steel production. The low porosity of the fused magnesia and the size of the periclase crystals considerably improve the corrosion resistance of the refractory lining. The refractory–grade material has a lower SiO_2 content and a higher CaO : SiO_2 ratio than the electrical-grade fused magnesia (i.e., for electrical heating, see Table 6).

In *special applications*, fused magnesia is also used as a molding material for precision casting, as a raw material for magnesium oxide ceramics and as single crystals [100] for optical windows and lenses.

4.6.4. Economic Aspects

The most important countries and companies producing fused magnesia are:

France	Pechiney Electrometallurgie
Germany	Hüls AG
United Kingdom	Universal Abrasives, Thermal Syndicate
Austria	Österreichisch Amerikanische Magnesit AG (refractory grades)
United States	Muscle Shoals Minerals, Tateho America (formerly CE-Minerals), Insultherm
Canada	Baymag (refractory grades)
Japan	Tateho Chemical Ind.

Further information on these manufacturers (e.g., production capacities) is given in [101].

The world consumption of fused magnesia for electrical heating (electrical grade) is estimated to be ca. 35 000–40 000 t/a. There are no figures available for refractory grade material because the grades of fused magnesia are not listed separately in the statistics.

5. Magnesium Sulfate

Magnesium sulfate does not occur in nature in anhydrous form. It is found in the form of hydrates and double salts in salt and potash deposits (e.g., in the Unites States, Germany, Russia, and the Ukraine). Magnesium sulfate also occurs in dissolved form in salt lakes.

5.1. Properties

The transformation temperatures of the stable magnesium sulfate hydrates and their solubilities in water are given in Table 7. Table 8 lists the molecular masses, the specific heats, refractive indices, and crystal classes of the anhydrous salt and some of the hydrates [102, 103]. Anhydrous magnesium sulfate can be obtained without decomposition by

Table 7. Stable magnesium sulfate hydrates

Hydrates	Transformation temperature, °C	Solubility, g $MgSO_4$/100 g H_2O
Ice–$MgSO_4 \cdot 12\ H_2O$	−3.9	22.0
$MgSO_4 \cdot 12\ H_2O$–$MgSO_4 \cdot 7\ H_2O$	1.8	27.2
$MgSO_4 \cdot 7\ H_2O$–$MgSO_4 \cdot 6\ H_2O$	48.1	49.5
$MgSO_4 \cdot 6\ H_2O$–$MgSO_4 \cdot H_2O$	67.5	56.6

Table 8. Properties of magnesium sulfate and its hydrates

Compound [CAS No.]	M_r	Specific heat, kJ kg^{-1} K^{-1}	Density, g/cm^3	Crystal class	Refractive index
$MgSO_4$ [*7487-88-9*]	120.37	0.800	2.66	orthorhombic	
$MgSO_4 \cdot H_2O$ [*14168-73-1*]	138.38	1.047	2.57	monoclinic	1.523, 1.535, 1.586
$MgSO_4 \cdot 2\ H_2O$ [*17830-05-6*]	156.40	1.124			
$MgSO_4 \cdot 4\ H_2O$ [*24378-31-2*]	192.43	1.305	2.01	monoclinic	1.490, 1.491, 1.497
$MgSO_4 \cdot 6\ H_2O$ [*13778-97-7*]	228.46	1.525	1.75		1.456, 1.453, 1.426
$MgSO_4 \cdot 7\ H_2O$ [*10034-99-8*]	246.48	1.546	1.68	orthorhombic	1.432, 1.4554, 1.4609

dehydrating the hydrates at 400–500 °C. At 700 °C, however, the compound decomposes to give magnesium oxide, sulfur dioxide, and oxygen. This accounts for the fact that the melting point of magnesium sulfate cannot be accurately determined, it is between 1120 and 1150 °C.

Magnesium sulfate is hygroscopic and absorbs water to form hydrates, finally the heptahydrate. Various hydrates can be obtained by stepwise dehydration of higher hydrates or by crystallization from aqueous solution. Hydrates of analogous composition can be produced by these two methods, but X-ray crystallography shows that their structures differ due to destruction of the crystal lattice during dehydration.

Five stable hydrates can be crystallized from aqueous solutions: $MgSO_4 \cdot 12\ H_2O$, $MgSO_4 \cdot 7\ H_2O$ (Epsom salt, epsomite, Bittersalz), $MgSO_4 \cdot 6\ H_2O$ (hexahydrate), $MgSO_4 \cdot 5/4\ H_2O$ (5/4 hydrate or synthetic kieserite) and $MgSO_4 \cdot H_2O$ (kieserite). Three metastable forms also occur: a second form of the heptahydrate, $MgSO_4 \cdot 5\ H_2O$ (allenite), and $MgSO_4 \cdot 4\ H_2O$ (leonardite). Stepwise dehydration yields hydrates with 1, 5/4, 2, 3, 4, and 6 molecules of water. The solubility of kieserite decreases with increasing temperature due to its negative heat of dissolution. Its dissolution rate increases with increasing temperature and decreasing pH.

The acid sulfates $MgSO_4 \cdot H_2SO_4$ and $MgSO_4 \cdot 3\ H_2SO_4$ crystallize when $MgSO_4$ is dissolved in sulfuric acid. Magnesium sulfate ammine hydrates ($MgSO_4 \cdot 3\ NH_3 \cdot 3\ H_2O$, $MgSO_4 \cdot 2\ NH_3 \cdot 4\ H_2O$, and $MgSO_4 \cdot 2\ NH_3 \cdot 2\ H_2O$) are produced when Epsom salt reacts with gaseous ammonia. On reaction with alkali, magnesium hydroxide precipitates. A precipitate of nesquehonite ($MgCO_3 \cdot 3\ H_2O$) is formed with sodium carbonate. After addition of sulfite, phosphate, or stearate solutions, insoluble magnesium sulfite, magnesium phosphate, or magnesium stearate precipitates. Magnesium chloride reacts with magnesium oxide to form oxychlorides; magnesium sulfate reacts in the same way to form oxysulfates ($MgSO_4 \cdot 3\ MgO \cdot 11\ H_2O$ and $MgSO_4 \cdot 5\ MgO \cdot 8\ H_2O$): it can therefore be used as a slower type of sulfate Sorel cement binder (see Sections 3.7, 4.4.3). Magnesium sulfate forms double salts with other salts, most of which also occur in nature:

$KCl \cdot MgSO_4 \cdot 3\ H_2O$ [*67145-93-1*] kainite
$K_2SO_4 \cdot 2\ MgSO_4$ [*13826-56-7*] langbeinite
$K_2SO_4 \cdot MgSO_4 \cdot 4\ H_2O$ [*15226-80-9*] leonite
$K_2SO_4 \cdot MgSO_4 \cdot 6\ H_2O$ [*15491-86-8*] schönite (picromerite)
$Na_2SO_4 \cdot MgSO_4 \cdot 4\ H_2O$ [*15083-77-9*] astrakanite (bloedite)
$6\ Na_2SO_4 \cdot 7\ MgSO_4 \cdot 15\ H_2O$ [*16633-52-6*] loeweite
$K_2SO_4 \cdot MgSO_4 \cdot 2\ CaSO_4 \cdot 2\ H_2O$ [*15278-29-2*] polyhalite

5.2. Occurrence and Raw Materials

The most important source of $MgSO_4$ in Germany is kieserite, which is a constituent of raw potash salt. In the United States, $MgSO_4$ is obtained from langbeinite and in Italy from kainite. The raw salts used by the potash industry in the Ukraine (Stebnik, Kalush) contain magnesium sulfate salts. Magnesium sulfate is also found in the brines of the Great Salt Lake and the Kara–Bogaz Gulf of the Caspian Sea and in other lakes, for example, the Dalagton Basin in China [104]. Seawater bitterns provide a further source (Epsom salt, see Section 5.3.2), the $MgSO_4$ is obtained by cooling. In many countries $MgSO_4$ is also produced from magnesite, dolomite, seawater magnesia, or other magnesium minerals.

5.3. Production

5.3.1. Production of Kieserite ($MgSO_4 \cdot H_2O$)

Production of Kieserite by Selective Dissolution of Residues from Potash Salt Production. The raw material for this process is a crude salt that contains kieserite (hard salt, carnallitite). The sodium chloride is selectively dissolved from residues of potash salt production. The filtered or centrifuged residues obtained during the production of potassium chloride are first washed with a partially saturated NaCl solution from a subsequent washing step. The NaCl in the residues is readily soluble and dissolves in the NaCl solution until it becomes saturated; the kieserite, however, remains largely undissolved due to its slow dissolution rate. Kieserite and residual NaCl are separated from the washing medium with a bucket conveyor or elevator. The kieserite–NaCl mixture is then washed with water in a second stage to dissolve any residual NaCl. Washing is performed in pipes lined with fused basalt that are several hundred meters long. The quantity of NaCl solution produced can be minimized by using the electrostatic production process (see below).

Ecological Aspects. The saturated NaCl solution contains 280–300 g NaCl per liter. Its disposal entails considerable ecological problems. Large quantities have been stored in jointed rock formations (platy dolomite) that are filled with slightly salted groundwater, but with only limited capacity [105].

Production of Kieserite by Flotation. Kieserite can be separated from rock salt by flotation. Prestaminol, oleic acid, fatty alcohol sulfates, or oxystearic acid are used as flotation agents for kieserite and anhydrite ($CaSO_4$) [106]. The kieserite can be further purified by separating it from the anhydrite and/or langbeinite. This is achieved by using oleic acid or a fatty acid amine as a flotation agent for the kieserite [107–109].

Electrostatic Separation of Kieserite. The electrostatic separation process (ESTA process) developed by Kali und Salz AG (Germany) [110] eliminates the formation of the large quantities of NaCl solution produced in the selective dissolution process (see above) [111, 112].

Kieserite can be electrostatically separated in one or two stages. In the one-stage process the raw salt is treated with aliphatic or a mixture of aliphatic and aromatic monocarboxylic acids that are combined with ammonium salts of aliphatic carboxylic acids, usually ammonium acetate. The salt is then heated at a defined relative humidity and electrostatically charged [113]. The kieserite becomes positively charged with respect to the potassium chloride and sodium chloride. Repeated separation of the kieserite deposited at the negative electrode yields $MgSO_4$ in a pure, marketable form.

In the two-stage process rock salt is first separated and then the highly pure kieserite. The raw salt is first treated with aromatic monocarboxylic acids, heated at a defined relative humidity, and then charged [114]. The rock salt becomes positively charged with respect to the potassium chloride and the kieserite. The rock salt is removed and the KCl–kieserite mixture separated at the positive electrode. The mixture is treated with aliphatic monocarboxylic acids, heated once again at a defined relative humidity, and charged [113]. The kieserite now becomes positive with respect to the potassium chloride. Repeated separation of the kieserite deposited at the negative electrode gives a pure, marketable product.

The potassium chloride fraction is still contaminated with kieserite and sodium chloride; it is used for the production of potassium salts [115, 116].

Production of Granulated Kieserite. After addition of 3 parts of a solution containing preferably 18 wt% Na_2SO_4 or 20 wt% Na_3PO_4 to 47 parts of kieserite, the product can be pressed in a moist state and then ground [117]. After addition of ammonium sulfate and hot magnesium sulfate solution, kieserite can also be granulated with a dish granulator [118]. Mono-, di-, or polysaccharides (usually lactose) are highly suitable additives for preparing these rolled granulates. The kieserite should have a broad particle size range; at least 20% of the particles must be less than 0.09 mm in diameter [119]. Addition of phosphate allows production of a rolled granulate of high strength even without drying [121, 122].

5.3.2. Production of Epsom Salt

Production of Epsom Salt from Kieserite. Kieserite is dissolved in hot water (90 °C) to give a clear solution. Dissolution is performed in single large vats with capacities of up to 500 m^3 that have a sieve base covered with coconut mats. The process has the disadvantage that the throughput is very low and the sieve plate gradually becomes encrusted with anhydrite and double salts. Kali und Salz AG therefore uses a multichamber cascade with slow stirring; the slurry density in the chambers is 70 wt%. Hot water is added from below and kieserite from above. The solution produced at the top of the cascade contains 400 g/L $MgSO_4$. At the bottom the residue containing anhydrite can be removed at intervals [123].

The hot concentrated solution is cooled in multistage vacuum coolers in which large crystals of Epsom salt are formed. After thickening, the Epsom salt is either filtered or centrifuged. The moist salt is dried in drum driers with air at ca. 50 °C.

Production of Epsom Salt from Magnesia, Magnesite, and Other Materials. Magnesium sulfate solution can be produced by treating seawater magnesia (MgO) or magnesite ($MgCO_3$) with sulfuric acid. Waste sulfuric acid contaminated with organic compounds may be used [124]. The resulting solution is clarified, evaporated to a density of 1.35–1.36 kg/L, and processed as described above. The product can be spray dried at 150 °C. Dolomite, olivine, serpentine, or peridotite can also be reacted with sulfuric acid. Magnesium sulfate can also be obtained by adding MgO to pyrite during roasting and subsequent extraction [125].

Waste serpentine from asbestos production can be reacted with ammonium sulfate above 250 °C and the reaction mixture extracted with water [126, 127]. The resulting solution is evaporated and the Epsom salt crystallizes.

Production of Epsom Salt from Boron- and Magnesium-Containing Raw Materials. The borate raw material containing SiO_2, Fe oxide, and Mg is decomposed with H_2SO_4. The solution is separated from the slurry. After crystallization, H_3BO_3 and $MgSO_4$ are separated by flotation [128].

Production of Epsom Salt from Seawater Bitterns or Salt Lake Brines. Pure Epsom salt can be obtained directly from seawater bitterns and from brines from salt lakes with a high $MgSO_4$ content (e.g., Kara–Bogaz Bay) by diluting with water and cooling to − 10 °C [129, 130].

Production of Epsom Salt from Langbeinite. Langbeinite ($K_2SO_4 \cdot 2\,MgSO_4$) is decomposed by treating with water for 6 h at 50–60 °C to produce $MgSO_4 \cdot 6\,H_2O$. On cooling the mother liquor to 20–35 °C the Epsom salt crystallizes [131].

Production of Epsom Salt from Magnesite or Dolomite and Gypsum. A mixture of ground dolomite and gypsum is treated with steam at 1.5–3.0 MPa and filtered to remove the $CaCO_3$ from the $MgSO_4$ solution [132]. Magnesite is calcined at 760–835 °C, hydrated in the presence of gypsum at 56–60 °C, and carbonized with carbon dioxide. The $MgSO_4$ solution is filtered to remove the $CaCO_3$ [133] and processed to obtain Epsom salt.

5.3.3. Production of Anhydrous Magnesium Sulfate

Anhydrous $MgSO_4$ does not occur in nature and can only be produced by dehydrating $MgSO_4$ hydrates. Kieserite is purified by removing anhydrite and langbeinite in a special electrostatic procedure. It is subsequently washed with water to remove residual chloride [134]; langbeinite and anhydrite may also be removed by flotation (Section 5.3.1). It is then dehydrated by heating at ca. 500 °C in a gas-heated calcination drum and finally cooled [135]. If Epsom salt is used as starting material instead of kieserite, six molecules of water of crystallization can be removed at < 70 °C and the remaining molecule at 500 °C [136].

5.4. Quality Specifications and Analysis

Magnesium sulfate is sold as kieserite (73 – 83 wt% $MgSO_4$), 99.9% pure Epsom salt (48.8% $MgSO_4$), and anhydrous $MgSO_4$

(> 98% $MgSO_4$). Epsom salt is marketed as a chemically pure product, as a clinical product that complies with pharmacopeia specifications, and as an analytical grade reagent. Maximum contamination limits for clinical grades vary according to the pharmacopeia: chloride 100–300 mg/kg, iron 20 mg/kg, and heavy metals 10 mg/kg. Maximum limits for arsenic, selenium, and other substances may also be specified.

The analytical procedures used for quality control are the same as those employed for magnesium chloride (see Section 3.5). Sulfate, however, is determined by a gravimetric method [137].

5.5. Transportation and Storage

Depending on the atmospheric humidity, Epsom salt can lose or absorb water during storage and thus tends to harden. It is therefore preferably supplied in Clupack valve sacks. Kieserite is transported in railroad cars and trucks, both in bulk and in valve paper sacks. Anhydrous $MgSO_4$ is also supplied in bulk and in sacks.

5.6. Uses

Large quantities of kieserite and Epsom salt are used in Germany for the production of potassium sulfate (→ Potassium Compounds). Kieserite is used in fertilizers, either as an additive in multinutrient fertilizers or directly to combat magnesium deficiency. In multinutrient fertilizers kieserite aids granulation and improves granulate properties. In cases of acute magnesium deficiency, Epsom salt is an especially effective foliar fertilizer.

Magnesium sulfate is a component of magnesium binders (Sorel cement, see Section 3.7) and can also be used as a binding agent in the production of magnesia bricks. Addition of $MgSO_4$ and $Al_2(SO_4)_3$ leads to the formation of spinels during firing of refractory products. It is also used to produce abrasive wheels. Decomposition of $MgSO_4$ in refractory products at > 1100 C results in the formation of sulfur dioxide and thus gives a light product. Magnesium sulfate is also used in the production of Portland cement from blast furnace slag and as an additive in gypsum-containing building materials.

Magnesium sulfate is employed in the sugar industry for refining the sugar liquor and for improving the sugar yield from molasses (Quentin procedure) [138]. It is utilized in fermentations for the production of amino acids, antibiotics, bakers' yeast, biomass, and citric acid. It increases the activity of glucose isomerase, the enzyme which converts glucose to fructose.

In the aluminum industry, a mixture of kieserite and calcium oxide or hydroxide is used to precipitate contaminants from aluminate solutions. Magnesium sulfate is also employed in fixative baths used to color aluminum surfaces by means of anodic oxidation. It is a component of the electrolytic baths used for chrome plating, zinc plating, copper plating, nickel plating, and the electroplating of tungsten, manganese, vanadium, cobalt, and cobalt alloys. It is also used in the phosphating of iron.

In the glass industry, sinters with a defined porosity can be obtained by incorporating $MgSO_4$ of a defined particle size and then leaching it. Magnesium sulfate is used as an additive in enamels and pigments. In the detergent industry it prevents gel formation during spray drying; magnesium silicate synthesized from $MgSO_4$ stabilizes peroxides and thus improves bleaching. Magnesium sulfate is also used as a make-up chemical in the sulfite pulping process and as a stabilizer for cellulose during oxygen bleaching of pulp. In the paper industry it is used in the preparation of paper for coating and in the deinking of recycled printed paper.

Other uses include treatment of waste and wastewater and soil stabilization. The compound is also used as an additive in drilling fluids, lead–acid batteries, oil and coal fuels, ore preparation, "isotonic" beverages, low-NaCl salt mixtures, cosmetics, bath salts, infusions, photographic developers, dye baths, and animal feeds.

In the plastics industry $MgSO_4$ is employed as a coagulant, polymerization catalyst, stabilizer, and flame retardant. In the petrochemical industry it is employed together with manganese chloride for cracking petroleum. Use of $MgSO_4$ in the leather industry makes the leather more supple, increases its weight, and removes calcium carbonate. Magnesium sulfate is also used as a heat storage medium and as a raw material for the production of magnesium aluminate, magnesium alumosilicate, and other magnesium compounds.

One of the important applications of Epsom salt is as a laxative.

5.7. Economic Aspects

China dominates the magnesium sulfate market, producing about two-thirds and consuming about one-third of worldwide production [80].

Kieserite is also produced and exported by K+S Kali GmbH (Kassel, Germany).

5.8. Toxicology and Occupational Health

Magnesium sulfate and its hydrates are not dangerous to human health when handled in accordance with the appropriate regulations. The LDL_0' (mouse, oral) for $MgSO_4$ is 5000 mg/kg. Magnesium sulfate is not listed in the EEC directive of May 5th, 1976 concerning the discharge of harzardous substances into water; it is classified as a nonhazardous substance in the EEC directive of June 12th, 1987 (WGK:O).

6. Organomagnesium Compounds

Organomagnesium compounds have been of outstanding importance in preparative organic chemistry ever since Victor Grignard (1900) discovered that they are readily prepared from organic halides and elemental magnesium in ethers [139]. In 1912 Grignard received the Nobel Prize for their discovery and for the development of their synthetic potential. As a divalent element, magnesium can bond with one (RMgX, X is usually a halogen, Grignard reagent), or two organic groups (R_2Mg); this bonding does not, however, satisfy the coordination requirements of magnesium (Section 6.1).

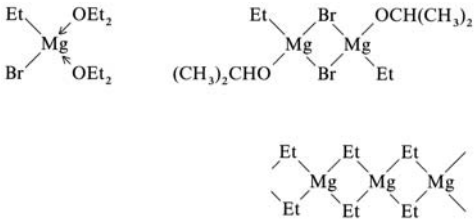
The production, structure, and properties of organomagnesium compounds have been summarized in several reviews [140–145]. Industrial use is generally limited to the production of fine chemicals, in which the upper limit is of the order of several tons.

Occasionally, they are used for the synthesis of (polymerization) catalysts [146, 147]. Their use for the production of organic lead derivatives (e.g., tetramethyl lead) is declining.

6.1. Properties

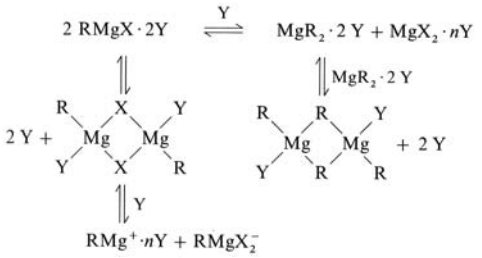
Compounds of the type RMgX or R_2Mg (R = aryl or alkyl) are generally colorless solids unless R is a resonance-stabilized, colored anion. With few exceptions (e.g., *sec*-dibutylmagnesium, dineopentylmagnesium, dicyclopentadienylmagnesium) they are nonvolatile; only the dialkyl compounds where $R > C_4H_9$ are soluble in hydrocarbons, especially aromatic hydrocarbons. Solubility, however, increases considerably in the presence of one or two equivalents of a coordinating ether, especially if intramolecular coordination occurs (i.e., by an ether function incorporated in the substituent R).

The structures of organomagnesium compounds in the solid state are being increasingly documented by crystal structure analysis. In the great majority of compounds, magnesium has a coordination number of four, which is generally attained by coordination with two Lewis bases, usually ethers (e.g., $RMgX \cdot 2\ Et_2O$); in the absence of a base, association takes place. The coordination sphere around the tetracoordinated magnesium is a distorted tetrahedron [141]. Association takes place by means of three-center, two-electron bonds, leading to the formation of dimeric or polymeric structures [148, 149]:



Organomagnesium compounds are thermally stable up to 100–170 °C, but at higher temperatures they can decompose explosively. They are extremely sensitive in air (i.e., in the presence of oxygen) and particularly toward water; they can sometimes ignite spontaneously. In ethereal solution the relatively heavy ether vapor protects them to some extent from exposure to air and thereby against decomposition. For small amounts or very pure preparations, an inert atmosphere of nitrogen or, preferably, argon, is recommended; in extreme cases an evacuated system should be employed.

For preparative purposes organomagnesium compounds are not isolated as such, but are used directly in solution following their preparation. Their structure in solution is basically the same as that in the crystal; nevertheless, a fundamental difference is that an equilibrium can exist between the various possible species, the position of which depends on the organic group R, the halogen X, the solvent Y, the concentration, and the temperature:



The most important of these equilibria is that between RMgX and R_2Mg, the so-called Schlenk equilibrium. It is strongly dependent on the solvent and lies, for example in diethyl ether, well to the left-hand side, whereas in tetrahydrofuran an almost statistical distribution exists (i.e., $R_2Mg : MgX_2 : 2\ RMgX = 1 : 1 : 2$).

Organomagnesium solutions can be quantitatively analyzed by various methods [150]. In addition to qualitative color tests [151], titrations can be used in which the active carbon–magnesium bond is either titrated directly with *sec*-butanol in the presence of a color indicator [152] or indirectly with hydrochloric acid after hydrolysis to give the free base, if necessary together with complexometric titration of the magnesium. Less convenient, but nevertheless reliable, is the formation and quantitative determination of derivatives, e.g., with carbon dioxide or chlorotrimethylstannane.

$$RCOOH \xleftarrow{CO_2,\,H^+} RMgX \xrightarrow{ClSn(CH_3)_3} RSn(CH_3)_3$$

Apart from their instability in air and their caustic basic properties, organomagnesium compounds and the precipitated inorganic magnesium salts produced during their use display no pronounced toxicity or side effects of environmental concern. The pyrophoric properties of the compounds and the high flammability of the ethereal solvents demand the usual protective measures.

6.2. Production

In practice the synthesis of Grignard reagents RMgX (X = halide) from organic halides RX and metallic magnesium is by far the most important preparative method. The Grignard reagents are used directly or indirectly as starting materials for the production of other (organometallic) compounds and also for the diorganylmagnesium compounds R_2Mg.

6.2.1. Grignard Reagents

The production of solutions of Grignard reagents (RMgX) according to the equation

$$R{-}X + Mg \xrightarrow{2\,Y} R{-}Mg{-}X \cdot 2Y$$

R = alkyl or aryl group
X = halogen
Y = solvent (usually diethyl ether or tetrahydrofuran)

presents few problems if suitable protective measures are observed [141, 153]. The fluorides react very sluggishly; the iodides are very reactive and side reactions, particularly Wurtz coupling, occur frequently. Side reactions also occur to a small extent with the chlorides and the bromides:

$$2\,R{-}X + Mg \rightarrow R{-}R + MgX_2$$

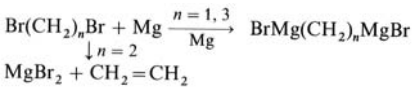
Organic bromides are most suitable for the production of Grignard reagents, but are frequently replaced in industrial applications by the cheaper chlorides.

The initiation of the reaction leading to formation of the Grignard reagent presents a considerable problem; it can be delayed or even fail as a result of reagent impurity or the low reactivity of the organic halide. Various strategies have been developed to overcome these difficulties [153], of which only a few will be mentioned here. Impurities causing delay in initiation can often be eliminated by the addition of a small amount of RMgX, possibly from an earlier batch. Most methods depend, however, on the activation of the magnesium, which is normally used in the form of turnings. Mechanical activation can be achieved by vigorous stirring or with ultrasound [154], chemical activation with numerous reagents (e.g., iodine, bromoethane, 1,2-dibromoethane, mercuric chloride, amalgam). A

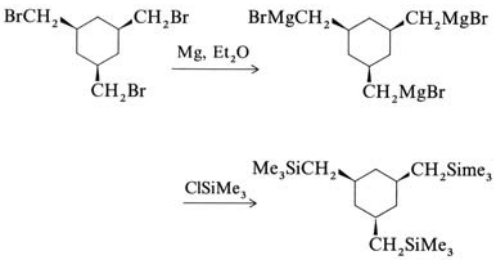
highly reactive, finely divided magnesium suspension can be obtained by (co-)sublimation of the metal (with the substrate), by reduction of magnesium chloride with potassium [155], or by heating anthracene–magnesium in tetrahydrofuran [156].

For reasons of cost and safety (inflammability, peroxide formation), the use of hydrocarbon solvents instead of the usual ethers is of interest. The synthesis of Grignard reagents in hydrocarbons (e.g., methylcyclohexane [146]) or an excess of the organic halide [157] unfortunately requires more severe conditions and leads to halogen-deficient products, whose reactivity often differs from that of the Grignard reagents synthesized in an ether solvent. Furthermore, solubility is also poor, so that suspensions must be used; however, the higher aliphatic halides (especially the iodides) give soluble organomagnesium compounds [158].

A relatively recent development is the synthesis of di-Grignard reagents, particularly those with only a few (1–3) carbon atoms between the two functional groups [159, 160]; they are of interest mainly in organometallic chemistry. Examples are the 1,1- or 1,3-di-Grignard reagents:

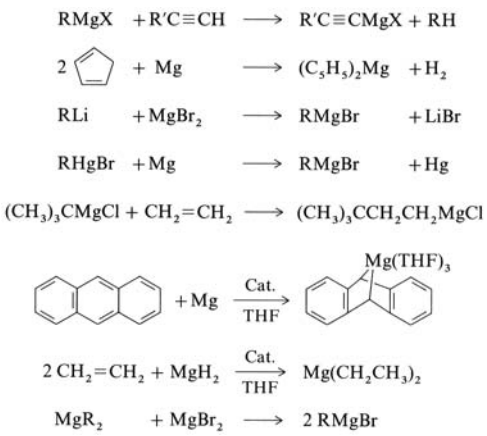


Even a tri-Grignard reagent has been produced (in 44% yield) [161]:



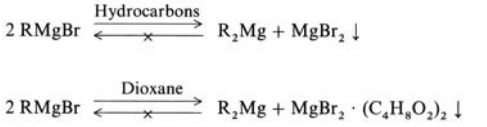
Finally, mention should be made of the Barbier reaction, i.e., the reaction of an organic halide with magnesium in the presence of the substrate (e.g., a ketone that is to be alkylated). This reaction was discovered by GRIGNARD's professor BARBIER and is the predecessor of the Grignard reaction. The Barbier reaction gives better yields when the Grignard reagents can only be isolated in poor yields, for example, with many allyl, benzyl-, or polyhalides [153, 154, 162, 163]: The method has been adapted for performance on a semi-industrial scale [164].

Other preparative methods have limited application and are only of special interest, e.g., the metallation of acidic hydrocarbons, metathesis with organolithium or organosodium compounds, metal exchange reaction with organomercury compounds, addition to unsaturated systems, or comproportionation of MgR_2 and $MgBr_2$ by the Schlenk equilibrium (in diethyl ether):



6.2.2. Diorganylmagnesium Compounds

Dialkyl- and diarylmagnesium compounds (R_2Mg) are generally of only limited practical importance. They are most frequently obtained from Grignard reagents by displacing the Schlenk equilibrium (see Section 6.1). In hydrocarbon solvents, the equilibrium already lies to the right. In diethyl ether, the equilibrium can be shifted by the addition of dioxane and precipitation of poorly soluble magnesium bromide–dioxanate [140, 141].



Very pure R_2Mg can be obtained by shaking diorganylmercury compounds for several days at room temperature in ether with an excess of magnesium.

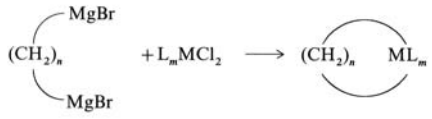
6.3. Uses

Organomagnesium compounds, especially Grignard reagents, have a wide range of application on account of their easy accessibility and their high reactivity. In organic synthesis they serve as nucleophiles for formation of carbon–carbon bonds. In organometallic chemistry they are used for the nucleophilic alkylation or arylation of metal salts; this reaction is suitable for the majority of main group and transition metals with a greater electronegativity than magnesium. In both areas the more reactive organolithium compounds are competing with the Grignard reagents. However, since the reactivity of Grignard reagents is usually high enough, they have lost little in popularity. This is also due to their ease of manipulation and their lower price, especially in industrial quantities.

The broad spectrum of reactions of Grignard reagents toward C=O, C=N, C≡N, C−X, C−H, and M−X groups can be illustrated by the following examples [140, 141, 143].

$$
\begin{aligned}
RMgX + R'CH{=}O &\rightarrow RR'CHOH\\
+ R'_2C{=}O &\rightarrow RR'_2CHOH\\
+ R'COOCH_2CH_3 &\rightarrow R_2R'COH\\
+ CO_2 &\rightarrow RCOOH\\
+ R'_2C{=}NR'' &\rightarrow RR'_2C{-}NHR''\\
+ R'C{\equiv}N &\xrightarrow{\text{Cat.}} RR'C{=}O\\
+ R'X &\rightarrow R{-}R'\\
+ R'H &\rightarrow R{-}H + R'MgX\\
+ MCl_n &\rightarrow MR_n
\end{aligned}
$$

(M = metal from group 3–15)



L = substituent or ligand $(n \geq 3)$

References

1 H. L. Pattinson, GB 9102, 1841.

2 W. Wetzenstein: "Limnische Huntit-Hydromagnesit-Magnesitlagerstätten in Macedonien/Nordgriechenland," *Miner. Deposita* **10** (1975) 129–140.

3 B. M. Coope: "Magnesia Markets. Fit for Optimism," *Ind. Miner. (London)*, Sept. 1989, 45–57.

General References

4 *Gmelin* **27** A, 57 ff.

5 H. Ginsberg, K. Wefers, *Die Metallischen Rohstoffe,* vol. 15: Aluminium und Magnesium, F. Enke Verlag, Stuttgart 1971.

Specific References

6 Dead Sea Works, DE-OS 2 532 022, 1975 (J. A. Epstein *et al.*).

7 B. Breiter, J. Steußloff, *Neue Bergbautech.* **16** (1986) no. 1, 23–27. Standard-Messo Verfahrenstechnik GmbH, DD 229 678, 1984 (R. Schmitz, K. Wamser, H. Zahalka).

8 C. Dähne, B. Maurer, *Neue Bergbautech.* **15** (1985) no. 8, 291–294.

9 Bechtel Internat. Corp., US 230 986, 1972; US 347 222, 1973. Kaliforschungsinstitut Sondershausen, DE-OS 1 467 293, 1964 (H. Hoppe); DE-AS 1 198 802, 1964 (H. Hoppe).

10 J. Kenat, *Symp. Salt (Proc.)* 2nd, vol. 2, 1965, 195–203.

11 Israel Chemicals Ltd., DE-OS 2 746 503, 1977 (A. Sadan, H. Pitnach).

12 E. E. Frolovskii, N. A. Alekseeva, L. B. Anikina, L. E. Bosiyak, *Khim. Promst. Ser. Kaliinaya Promst.* 1979, no. 3, 21–24; *Chem. Abstr.* **92** (1980) 78 938 f.

13 S. F. Estefan, *Hydrometallurgy* **10** (1983) 39–45.

14 Great Salt Lake Minerals and Chem. Corp., US 3 690 844, 1972 (U. Neitzel, H. G. Flint); P. Behrens, *Bull. Utah Geol. Miner. Surv.* **116** (1980) 223 –228.

15 Salzdetfurth AG, US 3 642 455, 1972 (B. J. Hahn, R. M. E. Reise, W. R. Raschka).

16 L. van den Assen, D. Buyze, H. P. Rogaar, *Rev. Metall. CIT*, April 1985, 295–305.

17 Kali und Salz, DE 2 513 947, 1975 (G. Budan, D. Kunze, W. Otto, I. Steingart).

18 H. Schnabel *et al*., DD 71 758, 1968.

19 Mines de Potasse d'Alsace, FR 8 104 391, 1981 (M. Bichara, M. Bodu, J. P. Koensgen, M. Meriaux, J.-P. Zimmermann).

20 E. S. Pereira, H. G. Riella: "Production of Pure Magnesium Chloride and Magnesium Oxide from Magnesite", *An. Assoc. Bras. Quim.* **44** (1995) no. 2, 15–18.

21 O. V. Lebedev: "Production of Magnesium Chloride", *Khim. Prom-st (Moscow) 1991* no. 2, 126; ISSN 0023–110X.

22 Aluminum Company of America, US 4 943 426, 1990 (L. E. Dastolfo, Jr., V. Cedro, III).

23 Kali und Salz, BASF, DE 3 119 968, 1981 (K. Wintermantel, D. Stockburger, A. Hollstein, D. Kunze, F. Werdelmann).

24 VEB Kombinat Kali, DD 211 105, 1982 (H. Scherzberg, W. Ulrich, R. Tober, W. Staufenbiel, M. Matern).

25 Mines de Potasse d'Alsace, DE 2 751 420, 1977 (A. Aubry, M. Bichara).

26 Nalco Chemical Company, DE-OS 3 010 755, 1980 (R. Allain, D. Braithwaite, J. P. Maniscalco). R. J. Allain, D. Braithwaite, *Proc. Annu. World Conf., Magnesium*, June 1979, 24–28.

27 Dow Chemical, US 3 282 642, 1966 (R. D. Goodenough, R. A. Gaska).
28 Kaiser Aluminium & Chem. Corp., US 3 181 930, 1931 (O. C. Olsen).
29 ESSO Res. & Eng. Co., US 3 347 625, 1965 (C. N. Kimberlin, Jr., W. F. Arey, Jr. F. J. Buchmann).
30 M. Nakayasu, Y. Suzukawa, W. Kobayashi: "Thermal Decomposition of $MgCl_2 \cdot NH_4Cl \cdot nNH_3$," *Denki Kagaku oyobi Kogyo Butsuri Kagaku* **51** (1983) no. 4, 396–402. *Chem. Abstr.* **99** (1983) 81 676 m.
31 J. T. May, V. E. Edlund, D. C. Seidel: "Dehydrating Magnesium Chloride by Double-Salt Decomposition". *Rep. Invest. U.S. Bur. Mines,* **RI 3277** (1978); *Chem. Abstr.* **89** (1978), 45 748 d.
32 Ruthner Engineering GmbH, DE-OS 2 638 123, 1976 (J. Jeney).
33 Ustav pro vyzkum rud, DE-OS 3 415 784, 1984 (V. Horák, V. Bumbálek, K. Matouš, F. Sehnálek).
34 Veitscher Magnesitwerke-AG, DE-OS 3 045 796, 1980 (M. Grill, H. Grohmann, E. Klein, A. Kullig).
35 *Kirk-Othmer*, 2nd ed., vol. 12, p. 672.
36 EEC Directive L 213/81–84, Method 5.1, Association of Official Analytical Chemists-Methods, 1984, no. 2.168, p. 30.
37 Association of Official Analytical Chemists-Methods (AOAC), 14th ed., 1984 no. 2.126–2.130 (AAS) and 2.141.
38 Association of Official Analytical Chemists-Methods (AOAC), 1984, Method no. 2.108–2.113.
39 Association of Official Analytical Chemists-Methods (AOAC), 1984, Method no. 2.173–2.176; Verband Deutscher Landwirtschaftlicher Untersuchungs- und Forschungsanstalten (Lufa): *Handbuch der landwirtschaftlichen Versuchs- und Untersuchungsmethodik (Methodenbuch)*, vol. 2: *Die Untersuchung von Düngemitteln,* Verlag J. Neumann-Neudamm, Melsungen 1973, 9_{62}– 9_{64}, Method 9.18.2.
40 J. Fries, H. Getrost: *Organische Reagenzien für die Spurenanalyse*, 3rd. ed., E. Merck, Darmstadt, Verlag Chemie, Weinheim, Germany 1966, p. 331 ff.
41 Lufa-Methodenbuch 923, Method 9.8.1.
42 Lufa-Methodenbuch 991–993, Method 9.28.2.
43 DIN 38405, part 17; H. Baron, *Fresenius Z. Anal. Chem.* **143** (1954) 339.
44 M. Zimmermann: *Fotometrische Metall- und Wasseranalysen*, Wiss. Verlagsanstalt, Stuttgart 1954.
45 E. Merck: *Organische Reagenzien für die Spurenanalyse*, 3rd ed., Verlag Chemie, Weinheim, Germany 1966, pp. 69–72.
46 Association of Official Analytical Chemists-Methods, Method no. 2.157–2.160, pp. 29/30.
47 R. Mathur, A. K. Misra, M. P. S. Chandrawa, *Res. Ind.* **31** (1986) June, 181–184. Kali und Salz, DE 3 230 962, 1982 (A. Singewald, K.-R. Löblich, H. Duyster). Kali und Salz, DE-OS 3 031 086, 1980 (H. Duyster, W. Prinzler, G. Nürnberger, K. Springer).
48 Saarbergwerke, DE-OS 2 204 281, 1972 (W. Schuhknecht, H. Kunz, H. G. Klinkner, G. Culmann); DE-OS 2 405 223, 1974 (H. G. Klinkner, G. Culmann).
49 Norsk Proco A/S, EP-A 0 241 103, 1987 (J. G. Berg, R. E. Smith-Johannsen).
50 M. C. Bell, *Feedstuffs* **50** (1978) no. 6, 24–25.
51 W. Pannekeet, *Chem. Ind. (London)* **21** (1983) 821 –824.
52 G. Gascho, *J. Fert. Issues* **3** (1986) no. 2, 62–65. Kali und Salz, DE-OS 3 432 327, 1984 (R. Wotschke, G. Budan, G. Kemmler).

General References

53 L. R. Duncan: "Today's Magnesia Industry," paper given at International Symposium on Refractories, Hangzhou, 1988.
54 E. Naujokat, D. v. Mallinckrodt, C. Zografou, *Keram. Zeitschr.* **35** (1983) 440–442.

Specific References

55 W. Zednicek: "Periklas. Ein bemerkenswertes und wesentliches Mineral in der Feuerfest-Keramik," *Radex Rundsch.* 1985, no. 4, 651–695.
56 V. Weiß: "Die Aufbereitung von Magnesit" in: *Handbuch der Keramik,* Verlag Schmid, Freiburg 1968.
57 H. Schmid: "Turkey's Salda Lake. A Genetic Model for Australias's Newly Discovered Magnesite Deposits," *Ind. Miner.* (*London)* August 1987, 19–31.
58 W. Wetzenstein: "Limnische Huntit-Hydromagnesit-Magnesitlagerstätten in Macedonien/Nordgriechenland," *Miner. Deposita* **10** (1975) 129–140.
59 W. Wetzenstein, D. Zachmann: "Sedimentäre magnesium-karbonatische Bildungen im Servia-Becken/Nordgriechenland," *Radex Rundsch.* 1977, no. 1, 29–49.
60 M. Coope: "Magnesia Markets. Fit for Optimism," *Ind. Miner. (London)* Sept. 1989, 45–57.
61 I. H. Howard-Smith: "Kunwarara Magnesite Deposit, Australia," Information Memorandum, Queensland Metals Corp., Toowong (Brisbane), Queensland, Australia, June 1989.
62 W. C. Gilpin, N. Heasman, *Chem. Ind. (London)*, 1977, 567–572.
63 J. C. Drum, S. Tangney, *Trans. J. Br. Ceramic Soc.* **77** (1978) no. 4, 10–14.
64 T. F. W. Barth, C. W. Correns, P. Eskola: *Die Entstehung der Gesteine*, Springer Verlag, Berlin 1960.
65 L. van den Assem: "Planning for New Industrial Minerals Projects–Magnesium Oxide from Brine in the Netherlands," *Ind. Miner. (London)*, January 1982, 35–42.
66 I. A. Epstein: "Utilization of the Dead Sea Minerals (a review)," *Hydrometallurgy* **2** (1976) 1–10.
67 H. Jedlicka: "Neue Anwendungen des Sprühröstverfahrens in der Chlorid-Naßmetallurgie," Schriftenreihe über Entwicklungen und Verfahren der Ruthner Industrieanlagen AG, Wien 1977, lecture Leningrad 1976.
68 H. K. Krivanec, W. F. Kladnig: "Herstellung von Oxidrohstoffen nach dem Andritz-Ruthner-Sprühröstverfahren," *TIZ* **112** (1988) no. 11, 762–769.
69 W. S. Ainscow: "Aufbereitung von Magnesit zu hochwertiger Sintermagnesia," *TIZ* **110** (1986) no. 6, 363–368. *Sulmag II the Sinter Magnesite Process*, Sulzer Brothers Ltd., Winterthur, Switzerland.

70 H. G. Schulte: "Polcal–An Economical Process for Calcining Magnesium Hydrate," *Ind. Miner. (London)*, November 1982, 37–47.

71 H. Priemer, G. Mörtl: "High Temperature Shaft Kiln for Large Crystal Periclase Materials," *Proceedings of International Symposium on Refractories*, Hangzhou 1988, 167–178.

72 Z. Foroglou *et al.*: "Refractory Raw Materials of Natural Origin and High Performance Refractories Based on such Products," *Proceedings of International Symposium on Refractories*, Hangzhou 1988, 204–213.

73 A. Eschner: "Die wärmetechnischen Eigenschaften von Magnesiasteinen für Niedertarifspeicherungen," Dissertation Fakultät für Bergbau, Hüttenwesen und Maschinenwesen der Techn. Universität Clausthal 1978.

74 "Magnesium Oxide Makes a Comeback," *Chem. Week*, January 1981, 43.

75 B. M. Coope: "Caustic Magnesia Markets–Agricultural Oversupply and Industrial Underdemand," *Ind. Miner. (London)* February 1981, 43–51.

76 I. Watson: "Minerals in Animal Feedstuffs–Plenty of Food for Thought," *Ind. Miner. (London)* April 1982, 71–91.

77 "Why Apply Magnesium with Potash?" *Phosphorus Potassium* **161** (1989) May–June, 20–28.

78 R. Schulze-Rettmer, T. Yawari: "Versuche mit dem Verfahren der Fällung von Magnesium-Ammonium-Phosphat (MAP) aus verschiedenen Abwässern," *Vom Wasser* **71** (1988) 41–54.

79 "Flammschutzmittel und Füllstoffe auf Basis hochreiner Magnesiumverbindungen," *Kunststoffe*, June 1988. *Japan Chemical Week*, 26th March 1987.

80 http://www.sricon sulting.com (accessed 25 June 2010).

81 R. C. Mackenzie: *The Differential Thermal Investigation of Clays*, Mineralogical Society (Clay Minerals Group), London 1957.

82 B. Brezny, *Ber. Dtsch. Keram. Ges.* **42** (1965) 308 –310.

83 I. S. Raeva, *Ogneupory* **24** (1983) no. 4, 31–34.

84 G. V. Samsonov (ed.): *The Oxide Handbook*, IFI/Plenum, Data Corporation, New York–Washington–London 1973, p. 185.

85 P. D. Johnson, *J. Am. Ceram. Soc.* **33** (1950) no. 5, 168–171.

86 S. P. Mitoff, *J. Chem. Phys.* **31** (1959) 1261–1269.

87 A. Cermak, *Silikattechnik* **18** (1967) no. 7, 219–223.

88 G. Schönfelder, *Elektrowärme Int. Ed. B* **41** (1983) B 5 214–221; Norton Co., DE-OS 2 333 601, 1973 (J. J. Scott).d

89 Norton Co., US 3 410 666, 1968 (J. J. Scott, N. C. Turnbull).

90 General Electric, DE-OS 1 592 116, 1966 (J. W. Rutter, E. M. Clausen).

91 Norton Co., US 2 280 515, 1939 (R. R. Ridgeway); Dynamit Nobel, DE 1 249 145, 1965 (H. R. Müller).

92 R. Czepek, *Elektrowärme Int.* **27** (1969) no. 12, 473 – 481.

93 Dynamit Nobel, DE-OS 1 592 089, 1967 (P. Hack, M. Neidhardt, H. Warmann).

94 General Electric, US 3 592 771, 1968 (W. Vedder, J. Schultz).

95 Norton Co., US 3 457 092, 1966 (R. O. Tervo).

96 Dynamit Nobel, DE 1 921 789, 1969 (M. Neidhardt, J. Schneider).

97 Dynamit Nobel, DE 2 363 790, 1973 (H. Clasen, K. Deneke).

98 Dynamit Nobel, DE 2 525 441, 1975 (G. E. Bockstiegel, M. Neidhardt, G. Rehfeld).

99 P. Bartha, F. Metz, K. H. Nitsch, *Stahl Eisen* **107** (1987) no. 13, 639–643.

100 A. Rabenau, *Chem. Ing. Techn.* **36** (1964) 542–545.

101 T. Power in E. M. Dickson (ed.): *Raw Materials for the Refractories Industry*: 2nd ed., Industrial Minerals, London 1986, pp. 163–177.

102 D'ANS-LAX: *Taschenbuch für Chemiker und Physiker*, vol. 1, Springer Verlag, Berlin–Heidelberg –New York 1967, pp. 417–418.

103 *Kirk-Othmer*, **14**, pp. 636–639.

104 "Features and Genesis of the Dalangtan Salt Deposit in the Qaidam Basin," *Z. Geol. Wiss.* **18** (1990) no. 9, 813–824. *Chem. Abstr.* 114 : 126079y.

105 Kali und Salz, DE 3 223 366, 1982 (U. Neitzel, H. Schroth, A. Hollstein, A. Singewald).

106 DD 25 546, 1958 (S. Mildner, L. Döhler).

107 VEB Kombinat Kali, DD 220 237, 1983 (L. Herrmann, H. Baldauf, W. Kramer, H. Schubert, F. Aurich).

108 Wintershall, DE 1 064 891, 1957 (G. Budan).

109 Duval Corp., US 4 045 335, 1977 (B. F. Adams, J. Jr.

110 Kali und Steinsalz, Bd. 12, Heft 2, Sept. 1996.

111 Kali und Salz, DE 3 637 226, 1986 (A. Singewald, U. Neitzel, G. Fricke).

112 Kali und Salz, DE 3 637 227, 1986 (A. Singewald, G. Fricke, H. Schroth).

113 Kali und Salz, DE 1 667 814, 1968 (G. Fricke, A. Singewald).

114 Kali und Salz, DE 1 792 120 (A. Singewald, G. Fricke).

115 G. Fricke, *Kali Steinsalz* **7** (1979) no. 12, 492.

116 A. Singewald, U. Neitzel, *Potash Technol. Min. Process. Maint. Transp. Occup. Health Saf. Environ.* [Int. Potash Technol. Conf. 1st] (1983) 589–595.

117 Kali und Salz, DE 3 148 404, 1981 (H. Zentgraf, H. Fetzer, A. Hollstein, G. Fricke).

118 Kali und Salz, DE 2 748 152, 1977 (G. Bruns).

119 Kali und Salz, DE 3 618 058, 1986 (K.-R. Löblich, G. Bruns, G. Peuschel).

120 Kali und Salz, DE 3 707 785, 1987 (K.-R. Löblich, G. Bruns, H. Zentgraf, E. Czaplinsky).

121 Kali und Salz, DE-PS 4 303 984 Cl, 1994 (K. R. Löblich, M. Schroeder, F. Schrumpf, R. Werthmann).

122 Kali und Salz, DE-PS 4 303 985 Cl, 1994 (N. Knoepfel, F. Schrumpf, T. Fleischer, R. Werthmann, R. Schad).

123 Kali und Salz, DE 2 512 246, 1975 (E. Höfling, H. Eberle, A. Hollstein, B. Hahn).

124 Wigton-Abbot Corp., US 2 754 175, 1956 (F. J. Hendel); *Chem. Abstr.* **77** (1972) 22 391 v; JP 7 116 245, 1971. *Chem. Abstr.* **89** (1978) 77 069 z; B. Mikhailov *et al.*, *Khim. Ind. (Sofia)* 1978, 50(3) 105–6.

125 Gypsum Industries Ltd., ZA 6 700 450, 1968 (N. Rudolf).
126 Société Nationale de l'Amiante, Thetford Mines, DE-OS 3 006 050, 1980 (J.-M. Lalancette); US 4 277 449, 1979.
127 *Chem. Abstr.* **95** (1989) 222 201 s; P. Bozadzhiev *et al.*, *God. Vissh. Khim-Tekhnol. Inst., Sofia*, **24** (1978) no. 2, 19–24.
128 SU 1 629 245 Al, 1991 (L. F. Kardashina *et al.*).
129 J. A. Fernandez-Lozano: "Recovery of Epsomite and Sylvite from Seawater Bittern by Crystallisation," *Fourth International Symposium on Salt*, Northern Ohio Geological Society, vol. 2, 1974, pp. 501–510.
130 *Chem. Abstr.* **85** (1976) 162 612 h.
131 IMCC, US 2 733 132, 1956. Sincat, FR 1 363 066, 1964.
132 *Chem. Abstr.* **89** (1978) 26 924 s; RO 59 695, 1971 (S. Kiss *et al.*).
133 SU 1 127 846, 1983 (N. P. Balandin); *Chem. Abstr.* **102** (1985) 134 380 g.
134 Kali und Salz, DE 3 334 665, 1983 (G. Fricke, I. Geisler).
135 R. Burmeister, *Bergakademie* **16** (1964) no. 9, 558 – 559.
136 "Verfahrenstechnische Alternativen bei der Kalzinierung von Kieserit unter Berücksichtigung der Optimierung des Energieeinsatzes," Techn. Universität Clausthal 1993, 101 S.
137 Association of Official Analytical Chemists-Methods (AOAC), Method no. 2.182–2.183, 1984, p. 32.
138 J. F. T. Oldfield *et al.*, *Int. Sugar J.* **82** (1979) 138 –143.
139 V. Grignard, *C. R. Hebd. Seances Acad. Sci.* **130** (1900) 1322.
140 K. Nützel in *Houben-Weyl*, vol. 13/2 a, pp. 47, 521; Science of Synthesis, vol. 7, 2004, p.503.
141 W. E. Lindsell in G. Wilkinson, F. G. A. Stone, E. W. Abel (eds.): *Comprehensive Organometallic Chemistry*, vol. 1, Pergamon Press, Oxford 1982, p. 155.
142 W. E. Lindsell in G. Wilkinson, F. G. A. Stone, E. W. Abel (eds.): *Comprehensive Organometallic Chemistry*, vol. II, Pergamon, Oxford 1995, p. 57.
143 Ch. Elschenbroich, A. Salzer: *Organometallics*, 2nd ed., VCH Verlagsgesellschaft, Weinheim, Germany 1992, p. 40.
144 G. S. Silverman, P. E. Rakita (eds.): *Handbook of Grignard Reagents*, Marcel Dekker, New York 1996.
145 H. G. Richey (ed.): *Grignard Reagents*, John Wiley & Sons, Chichester 1999.
146 D. Bryce-Smith, E. T. Blues, *Org. Synth.* **47** (1967) 113.
147 Monsanto, US 4 438 034, 1984 (A. Y. Garner); *Chem. Abstr.* **101** (1984) 24 119 a.
148 A. L. Spek, P. Voorbergen, G. Schat, C. Blomberg, F. Bickelhaupt, *J. Organomet. Chem.* **77** (1974) 147.
149 E. Weiss, *J. Organomet. Chem.* **4** (1965) 101.
150 C. Blomberg: "Magnesium" in F. Korte (ed.): *Methodicum Chimicum,* vol. 7, Thieme Verlag, Stuttgart 1976, p. 79.
151 H. Gilman, F. Schulze, *J. Am. Chem. Soc.* **47** (1925) 2002.
152 S. C. Watson, J. F. Eastham, *J. Organomet. Chem.* **9** (1967) 165.
153 F. Bickelhaupt in A. P. Hagen (ed.): *Inorganic Reactions and Methods*, vol. 10, VCH Publishers, New York 1989, p. 237.
154 C. Einhorn, J. Einhorn, J. L. Luche, *Synthesis* 1989, 787.
155 R. D. Rieke, S. E. Bales, P. M. Hudnall, G. S. Poindexter, *Org. Synth.* **59** (1980) 85.
156 B. Bogdanović, *Angew. Chem. Int. Ed. Engl.* **25** (1985) 262.
157 Oxon Italia S. p. A., DE-OS 2 625 216, 1976; *Chem. Abstr.* **86** (1977) 140 234.
158 W. N. Smith, Jr., *J. Organomet. Chem.* **64** (1974) 25.
159 F. Bickelhaupt, *Angew. Chem.* **99** (1987) 1020.
160 F. Bickelhaupt, *Chem. Soc. Rev.* **28** (1999) 17, and references cited therein.
161 Ph. Boudjouk, R. Sooriyakumaran, C. A. Kapfer, *J. Organomet. Chem.* **281** (1985) C 21.
162 C. Blomberg, F. A. Hartog, *Synthesis* 1977, 18.
163 C. Blomberg: "The Barbier Reaction and Related One-Step Processes," in K. Hafner, J. M. Lehn, C. W. Rees, P. von Ragué Schleyer, B. M. Trost, R. Zahradnik (eds.): *Reactivity and Structure Concepts in Organic Chemistry*, vol. 31, Springer-Verlag, Berlin Heidelberg 1993.
164 B. A. Klokov, *Main Group Metal Chemistry* **22** (1999) 1.

Further Reading

R. Abbaschian: *Physical Metallurgy Principles*, 4th ed., Thomson/Cengage, Mason OH 2008.
D. A. Brandt, J. C. Warner: *Metallurgy Fundamentals*, 5th ed., Goodheart-Willcox, Tinley Park, IL 2009.
P. Enghag: *Encyclopedia of the Elements*, Wiley-VCH, Weinheim 2004.
K. U. Kainer (ed.): *Magnesium*, Wiley-VCH, Weinheim 2010.
D. A. Kramer: *Magnesium Compounds*, "Kirk Othmer Encyclopedia of Chemical Technology", 5th edition, vol. 15, p. 381–433, John Wiley & Sons, Hoboken, NJ, 2005, online: DOI: 10.1002/0471238961.1301071410010311.a01.pub3.
E. J. Teichert: *Introduction to Ferrous Metallurgy*, Maurois Press 2008.
G. E. Totten, D. S. MacKenzie (eds.): *Handbook of Aluminum*, Dekker, New York 2003.